Supplement to "Beyond CTI Reporting: Evaluating LLM-Compiled Threat Intelligence as Adversary Behavior Models"

Jack Moody Iain Cruickshank

All numbers use the paper's adopted settings: exact hypergeometric chance, 10,000 resamples for every bootstrap (seed 20261105), and block resampling with blocks of 2, 3 or 5 consecutive transitions kept within one plan scenario. No model calls were made for any table here; every table is recomputed from the cached extractions. Kernel codes: R = report-conditioned, B = no report (actor name and aliases only), ATTK = ATT&CK-derived kernel. F = Claude Opus 5.5, G = GPT-6 Astra (API models; means of five runs), Q = Qwen3 14B, M = Ministral 3 14B (temperature 0), Q5 and M5 = the same models over five runs at default sampling, M3 = Ministral 3 3B.

1 Evidence filtering

The main analysis counts every schema-valid edge, whether or not its evidence quote matches the source. Here the report-conditioned kernels are rebuilt without edges whose quotes fail the fragment-aware check (the check described in the paper) or a strict whole-span check. B and ATTK kernels are unaffected. Each table shows the adopted value beside the filtered one.

Table S1: Share of report-conditioned edges, and of count units (edge × report support, summed over samples), that rest only on evidence quotes failing the fragment-aware or the strict check.

Model	Actor	samples	edges	edges only failing (fragment)	edges only failing (strict)	count units	units only failing (fragment)	units only failing (strict)
Claude Opus	APT29	5	219	0.0%	2.3%	317	0.3%	5.7%
Claude Opus	OilRig	5	218	0.0%	0.5%	328	0.0%	1.8%
Claude Opus	Sandworm	5	164	0.0%	6.1%	211	0.0%	10.9%
Claude Opus	menuPass	5	110	0.0%	2.7%	159	0.0%	4.4%
GPT-6 Astra	APT29	5	273	0.0%	0.0%	382	0.0%	0.0%
GPT-6 Astra	OilRig	5	241	0.0%	0.0%	411	0.0%	0.0%
GPT-6 Astra	Sandworm	5	191	0.0%	0.0%	258	0.0%	0.0%
GPT-6 Astra	menuPass	5	147	0.0%	0.0%	220	0.0%	0.0%
Qwen 14B	APT29	1	39	12.8%	12.8%	59	18.6%	18.6%
Qwen 14B	OilRig	1	24	12.5%	12.5%	41	14.6%	14.6%
Qwen 14B	Sandworm	1	32	12.5%	12.5%	40	12.5%	12.5%
Qwen 14B	menuPass	1	31	12.9%	16.1%	44	18.2%	20.5%
Qwen 14B sampled	APT29	5	224	1.8%	2.7%	371	5.1%	8.1%
Qwen 14B sampled	OilRig	5	131	9.9%	12.2%	225	12.9%	16.0%
Qwen 14B sampled	Sandworm	5	187	33.2%	34.2%	242	36.8%	37.6%
Qwen 14B sampled	menuPass	5	155	52.3%	56.1%	221	52.0%	57.0%
Ministral 14B	APT29	1	32	31.2%	31.2%	45	35.6%	35.6%
Ministral 14B	OilRig	1	33	9.1%	24.2%	47	19.1%	29.8%
Ministral 14B	Sandworm	1	15	6.7%	6.7%	20	25.0%	25.0%
Ministral 14B	menuPass	1	24	16.7%	16.7%	33	27.3%	27.3%
Ministral 14B sampled	APT29	5	207	14.5%	15.0%	321	18.4%	20.9%
Ministral 14B sampled	OilRig	5	165	20.6%	26.7%	246	27.2%	32.9%
Ministral 14B sampled	Sandworm	5	91	24.2%	25.3%	110	34.5%	35.5%
Ministral 14B sampled	menuPass	5	123	33.3%	35.8%	156	37.2%	39.1%
Ministral 3B	APT29	1	32	43.8%	43.8%	38	47.4%	47.4%
Ministral 3B	OilRig	1	34	55.9%	55.9%	50	62.0%	62.0%
Ministral 3B	Sandworm	1	16	56.2%	68.8%	16	56.2%	68.8%
Ministral 3B	menuPass	1	29	51.7%	58.6%	37	59.5%	64.9%
Ministral 3B sampled	APT29	5	141	35.5%	37.6%	182	42.3%	44.0%
Ministral 3B sampled	OilRig	5	163	54.0%	56.4%	242	62.8%	64.9%
Ministral 3B sampled	Sandworm	5	66	57.6%	60.6%	71	60.6%	63.4%
Ministral 3B sampled	menuPass	5	151	45.7%	54.3%	204	54.4%	64.2%

Table S2: Fragment-aware evidence filter: support recovery (cf. main Table 2). Edge-F1, F1 ÷ chance (exact), and percentile against chance, adopted vs. filtered.

Kernel	Actor	GT edges	F1 adopted	F1	F1/chance adopted	F1/chance	pct adopted	pct
R-F	APT29	28	0.371	0.371	+1.99	+1.99	99.8	99.8
R-F	OilRig	15	0.396	0.396	+3.23	+3.23	100.0	100.0
R-F	Sandworm	21	0.372	0.372	+2.65	+2.65	99.9	99.9
R-F	menuPass	6	0.300	0.300	+5.78	+5.78	99.9	99.9
R-G	APT29	28	0.450	0.450	+2.21	+2.21	100.0	100.0
R-G	OilRig	15	0.341	0.341	+2.72	+2.72	100.0	100.0
R-G	Sandworm	21	0.331	0.331	+2.22	+2.22	99.7	99.7
R-G	menuPass	6	0.181	0.181	+3.30	+3.30	97.5	97.5
R-Q	APT29	28	0.358	0.387	+2.00	+2.29	99.7	99.9
R-Q	OilRig	15	0.205	0.222	+2.02	+2.31	92.6	95.3
R-Q	Sandworm	21	0.340	0.327	+2.44	+2.48	99.8	99.6
R-Q	menuPass	6	0.270	0.303	+4.89	+5.62	100.0	100.0
R-Q5	APT29	28	0.359	0.357	+1.98	+1.99	97.5	97.7
R-Q5	OilRig	15	0.199	0.203	+1.88	+1.98	83.0	83.2
R-Q5	Sandworm	21	0.236	0.244	+1.71	+1.98	86.6	86.2
R-Q5	menuPass	6	0.239	0.375	+4.44	+8.17	99.5	99.8
R-M	APT29	28	0.333	0.320	+2.03	+2.36	99.4	99.5
R-M	OilRig	15	0.375	0.356	+3.31	+3.24	100.0	100.0
R-M	Sandworm	21	0.167	0.171	+1.73	+1.86	84.1	86.5
R-M	menuPass	6	0.267	0.308	+5.06	+6.07	99.8	99.9
R-M5	APT29	28	0.311	0.289	+1.71	+1.70	96.0	92.8
R-M5	OilRig	15	0.250	0.235	+2.21	+2.29	97.0	94.5
R-M5	Sandworm	21	0.265	0.208	+2.48	+2.33	96.0	92.4
R-M5	menuPass	6	0.290	0.282	+5.50	+5.97	99.6	98.4
R-M3	APT29	28	0.133	0.130	+0.81	+1.08	32.6	58.1
R-M3	OilRig	15	0.204	0.133	+1.78	+1.62	91.9	76.6
R-M3	Sandworm	21	0.162	0.071	+1.62	+1.24	81.5	61.5
R-M3	menuPass	6	0.229	0.300	+4.18	+6.50	99.7	99.7

Table S3: Fragment-aware evidence filter: information gain over uniform (cf. main Table 3), with i.i.d. 95% intervals (10,000 resamples of plan transitions), status, and segment-block marks (*), adopted vs. filtered.

Kernel	Plan	IG adopted	IG	i.i.d. 95%	status adopted	status	* adopted	*
R-F	APT29	+.11	+.11	[-0.13, +0.37]	spans	spans		
R-F	OilRig	+.62	+.62	[+0.17, +1.04]	above	above		
R-F	Sandworm	+.31	+.31	[-0.05, +0.68]	spans	spans		
R-F	menuPass	+.91	+.91	[+0.33, +1.49]	above	above		
R-F	Attr.	3/4	3/4					
R-G	APT29	+.30	+.30	[+0.05, +0.55]	above	above		
R-G	OilRig	+.55	+.55	[-0.01, +1.08]	spans	spans	*	*
R-G	Sandworm	+.18	+.18	[-0.15, +0.52]	spans	spans		
R-G	menuPass	+.39	+.39	[-0.31, +1.13]	spans	spans	*	*
R-G	Attr.	3/4	3/4					
R-Q	APT29	+.12	+.21	[-0.02, +0.45]	spans	spans		*
R-Q	OilRig	-.17	-.12	[-0.43, +0.22]	spans	spans	*	
R-Q	Sandworm	+.17	+.15	[-0.14, +0.45]	spans	spans		
R-Q	menuPass	+1.00	+.87	[+0.52, +1.09]	above	above		
R-Q	Attr.	1/4	2/4					
R-Q5	APT29	+.10	+.09	[-0.14, +0.33]	spans	spans		
R-Q5	OilRig	-.21	-.17	[-0.38, +0.06]	spans	spans	*	
R-Q5	Sandworm	-.02	+.02	[-0.13, +0.19]	spans	spans		
R-Q5	menuPass	+.86	+.81	[+0.22, +1.35]	above	above		
R-Q5	Attr.	1/4	3/4					
R-M	APT29	+.16	+.14	[-0.08, +0.38]	spans	spans		
R-M	OilRig	+.37	+.24	[-0.10, +0.59]	above	spans		*
R-M	Sandworm	-.04	-.02	[-0.22, +0.23]	spans	spans		
R-M	menuPass	+1.04	+.93	[+0.30, +1.54]	above	above		
R-M	Attr.	3/4	3/4					
R-M5	APT29	+.13	+.09	[-0.13, +0.31]	spans	spans		
R-M5	OilRig	+.07	+.05	[-0.26, +0.38]	spans	spans		
R-M5	Sandworm	+.06	+.01	[-0.16, +0.21]	spans	spans		
R-M5	menuPass	+1.07	+.68	[+0.18, +1.21]	above	above		
R-M5	Attr.	2/4	2/4					
R-M3	APT29	-.24	-.11	[-0.26, +0.06]	below	spans		
R-M3	OilRig	+.08	+.01	[-0.25, +0.30]	spans	spans		
R-M3	Sandworm	+.01	+.01	[-0.08, +0.14]	spans	spans		
R-M3	menuPass	+.89	+.66	[+0.20, +1.15]	above	above		
R-M3	Attr.	0/4	1/4					

Table S4: Fragment-aware evidence filter: R−B gaps for every model (cf. main Table 5 and Table S18), with report-bootstrap (10,000) and leave-one-report-out jackknife intervals for ΔF1 and Δenrichment, and the paired transition-level i.i.d. interval for ΔIG.

Model	Actor	ΔF1 adopted	ΔF1	boot	jackknife	clear adopted	clear	Δenr adopted	Δenr	boot	jackknife	clear adopted	clear	ΔIG adopted	ΔIG	i.i.d.	status adopted	status
Claude Opus	APT29	+0.137	+0.137	[-0.098, +0.178]	[-0.066, +0.340]	not clear	not clear	+0.03	+0.03	[-0.85, +0.46]	[-0.87, +0.93]	not clear	not clear	+0.06	+0.06	[-0.15, +0.27]	spans	spans
Claude Opus	OilRig	+0.280	+0.280	[+0.185, +0.329]	[+0.181, +0.380]	above	above	+1.86	+1.86	[+1.29, +2.98]	[+0.42, +3.30]	above	above	+0.61	+0.61	[+0.18, +1.02]	above	above
Claude Opus	Sandworm	+0.189	+0.189	[-0.012, +0.255]	[-0.009, +0.386]	not clear	not clear	+0.80	+0.80	[-0.05, +1.79]	[-0.30, +1.90]	not clear	not clear	+0.33	+0.33	[-0.07, +0.72]	spans	spans
Claude Opus	menuPass	-0.195	-0.195	[-0.491, -0.062]	[-0.544, +0.154]	not clear	not clear	-5.08	-5.08	[-10.73, -1.52]	[-11.98, +1.81]	not clear	not clear	-0.09	-0.09	[-0.51, +0.26]	spans	spans
GPT-6 Astra	APT29	+0.273	+0.273	[+0.023, +0.342]	[+0.021, +0.526]	above	above	+0.56	+0.56	[-0.41, +1.33]	[-0.51, +1.64]	not clear	not clear	+0.30	+0.30	[+0.06, +0.53]	above	above
GPT-6 Astra	OilRig	+0.193	+0.193	[+0.115, +0.297]	[+0.098, +0.288]	above	above	+0.34	+0.34	[-0.56, +2.16]	[-0.80, +1.49]	not clear	not clear	+0.50	+0.50	[-0.02, +1.00]	spans	spans
GPT-6 Astra	Sandworm	+0.150	+0.150	[-0.092, +0.217]	[-0.280, +0.580]	not clear	not clear	-0.22	-0.22	[-1.69, +0.64]	[-2.73, +2.28]	not clear	not clear	+0.17	+0.17	[-0.14, +0.46]	spans	spans
GPT-6 Astra	menuPass	-0.143	-0.143	[-0.307, -0.082]	[-0.223, -0.062]	below	below	-5.57	-5.57	[-8.85, -3.47]	[-6.87, -4.26]	below	below	-0.01	-0.01	[-0.92, +0.74]	spans	spans
Qwen 14B	APT29	+0.158	+0.187	[-0.042, +0.221]	[-0.194, +0.568]	not clear	not clear	-0.17	+0.13	[-0.46, +0.62]	[-1.39, +1.65]	not clear	not clear	+0.02	+0.11	[-0.04, +0.27]	spans	spans
Qwen 14B	OilRig	+0.134	+0.151	[+0.003, +0.187]	[-0.068, +0.369]	not clear	not clear	+1.09	+1.38	[+0.08, +2.10]	[-0.60, +3.35]	not clear	not clear	-0.06	-0.01	[-0.38, +0.37]	spans	spans
Qwen 14B	Sandworm	+0.222	+0.209	[-0.044, +0.216]	[-0.113, +0.531]	not clear	not clear	+1.10	+1.14	[+0.11, +1.30]	[-0.04, +2.32]	above	not clear	+0.25	+0.22	[-0.18, +0.60]	spans	spans
Qwen 14B	menuPass	+0.060	+0.093	[-0.211, +0.093]	[-0.326, +0.511]	not clear	not clear	+0.23	+0.95	[-4.67, +0.95]	[-5.86, +7.76]	not clear	not clear	+0.68	+0.55	[-0.05, +1.10]	above	spans
Qwen 14B sampled	APT29	+0.160	+0.158	[-0.018, +0.197]	[-0.015, +0.332]	not clear	not clear	-0.15	-0.15	[-0.90, +0.70]	[-1.13, +0.84]	not clear	not clear	+0.00	+0.00	[-0.17, +0.18]	spans	spans
Qwen 14B sampled	OilRig	+0.154	+0.158	[-0.037, +0.256]	[-0.155, +0.470]	not clear	not clear	+1.23	+1.34	[-0.59, +2.36]	[-1.53, +4.21]	not clear	not clear	-0.10	-0.06	[-0.29, +0.18]	spans	spans
Qwen 14B sampled	Sandworm	+0.130	+0.138	[-0.018, +0.181]	[+0.058, +0.217]	not clear	not clear	+0.49	+0.77	[-0.25, +1.37]	[+0.23, +1.31]	not clear	not clear	+0.06	+0.10	[-0.06, +0.26]	spans	spans
Qwen 14B sampled	menuPass	+0.019	+0.155	[-0.184, +0.237]	[-0.324, +0.634]	not clear	not clear	-0.54	+3.19	[-3.76, +6.67]	[-7.32, +13.69]	not clear	not clear	+0.51	+0.46	[+0.02, +0.94]	above	above
Ministral 14B	APT29	+0.233	+0.220	[-0.005, +0.264]	[-0.042, +0.482]	not clear	not clear	+0.95	+1.28	[-0.15, +2.17]	[-0.78, +3.34]	not clear	not clear	+0.17	+0.16	[-0.06, +0.39]	spans	spans
Ministral 14B	OilRig	+0.375	+0.356	[+0.143, +0.410]	[-0.040, +0.752]	above	not clear	+3.31	+3.24	[+1.87, +4.04]	[+0.35, +6.12]	above	above	+0.56	+0.43	[+0.06, +0.81]	above	above
Ministral 14B	Sandworm	+0.049	+0.054	[-0.118, +0.064]	[-0.057, +0.164]	not clear	not clear	+0.40	+0.52	[-1.33, +3.00]	[-1.88, +2.93]	not clear	not clear	-0.03	-0.01	[-0.34, +0.32]	spans	spans
Ministral 14B	menuPass	+0.044	+0.085	[-0.068, +0.159]	[-0.068, +0.239]	not clear	not clear	+0.00	+1.01	[-0.72, +3.61]	[-1.00, +3.02]	not clear	not clear	+0.72	+0.61	[-0.38, +1.47]	spans	spans
Ministral 14B sampled	APT29	+0.143	+0.121	[-0.077, +0.187]	[-0.059, +0.301]	not clear	not clear	-0.08	-0.08	[-1.07, +0.64]	[-0.91, +0.75]	not clear	not clear	+0.12	+0.07	[-0.09, +0.24]	spans	spans
Ministral 14B sampled	OilRig	+0.234	+0.220	[+0.134, +0.257]	[+0.184, +0.255]	above	above	+1.99	+2.06	[+1.39, +3.25]	[+1.17, +2.96]	above	above	+0.26	+0.24	[-0.06, +0.59]	spans	spans
Ministral 14B sampled	Sandworm	+0.161	+0.104	[-0.104, +0.153]	[-0.116, +0.323]	not clear	not clear	+1.04	+0.89	[-1.30, +1.97]	[-0.16, +1.95]	not clear	not clear	+0.07	+0.02	[-0.26, +0.30]	spans	spans
Ministral 14B sampled	menuPass	+0.025	+0.017	[-0.223, +0.108]	[-0.153, +0.187]	not clear	not clear	-0.54	-0.07	[-4.38, +2.21]	[-1.00, +0.86]	not clear	not clear	+0.64	+0.25	[-0.28, +0.97]	above	spans
Ministral 3B	APT29	+0.133	+0.130	[+0.000, +0.143]	[-0.047, +0.308]	not clear	not clear	–	–	–	–	undefined	undefined	-0.24	-0.11	[-0.26, +0.06]	below	spans
Ministral 3B	OilRig	+0.204	+0.133	[+0.000, +0.160]	[-0.061, +0.328]	above	not clear	–	–	–	–	undefined	undefined	+0.08	+0.01	[-0.25, +0.30]	spans	spans
Ministral 3B	Sandworm	+0.162	+0.071	[+0.000, +0.083]	[-0.088, +0.231]	above	not clear	–	–	–	–	undefined	undefined	+0.01	+0.01	[-0.08, +0.14]	spans	spans
Ministral 3B	menuPass	+0.229	+0.300	[+0.000, +0.429]	[-0.212, +0.812]	not clear	not clear	–	–	–	–	undefined	undefined	+0.89	+0.66	[+0.20, +1.15]	above	above

Table S5: Strict whole-span evidence filter: support recovery (cf. main Table 2).

Kernel	Actor	GT edges	F1 adopted	F1	F1/chance adopted	F1/chance	pct adopted	pct
R-F	APT29	28	0.371	0.360	+1.99	+1.95	99.8	99.6
R-F	OilRig	15	0.396	0.391	+3.23	+3.19	100.0	100.0
R-F	Sandworm	21	0.372	0.386	+2.65	+2.82	99.9	100.0
R-F	menuPass	6	0.300	0.306	+5.78	+5.94	99.9	99.9
R-G	APT29	28	0.450	0.450	+2.21	+2.21	100.0	100.0
R-G	OilRig	15	0.341	0.341	+2.72	+2.72	100.0	100.0
R-G	Sandworm	21	0.331	0.331	+2.22	+2.22	99.7	99.7
R-G	menuPass	6	0.181	0.181	+3.30	+3.30	97.5	97.5
R-Q	APT29	28	0.358	0.387	+2.00	+2.29	99.7	99.9
R-Q	OilRig	15	0.205	0.222	+2.02	+2.31	92.6	95.3
R-Q	Sandworm	21	0.340	0.327	+2.44	+2.48	99.8	99.6
R-Q	menuPass	6	0.270	0.312	+4.89	+5.83	100.0	100.0
R-Q5	APT29	28	0.359	0.360	+1.98	+2.03	97.5	97.7
R-Q5	OilRig	15	0.199	0.199	+1.88	+1.96	83.0	83.6
R-Q5	Sandworm	21	0.236	0.236	+1.71	+1.92	86.6	84.2
R-Q5	menuPass	6	0.239	0.397	+4.44	+8.87	99.5	99.9
R-M	APT29	28	0.333	0.320	+2.03	+2.36	99.4	99.5
R-M	OilRig	15	0.375	0.350	+3.31	+3.40	100.0	99.9
R-M	Sandworm	21	0.167	0.171	+1.73	+1.86	84.1	86.5
R-M	menuPass	6	0.267	0.308	+5.06	+6.07	99.8	99.9
R-M5	APT29	28	0.311	0.290	+1.71	+1.71	96.0	93.3
R-M5	OilRig	15	0.250	0.248	+2.21	+2.50	97.0	95.8
R-M5	Sandworm	21	0.265	0.209	+2.48	+2.35	96.0	92.9
R-M5	menuPass	6	0.290	0.287	+5.50	+6.10	99.6	98.7
R-M3	APT29	28	0.133	0.130	+0.81	+1.08	32.6	58.1
R-M3	OilRig	15	0.204	0.133	+1.78	+1.62	91.9	76.6
R-M3	Sandworm	21	0.162	0.077	+1.62	+1.73	81.5	71.8
R-M3	menuPass	6	0.229	0.222	+4.18	+5.06	99.7	97.2

Table S6: Strict whole-span evidence filter: information gain (cf. main Table 3).

Kernel	Plan	IG adopted	IG	i.i.d. 95%	status adopted	status	* adopted	*
R-F	APT29	+.11	+.08	[-0.16, +0.34]	spans	spans		
R-F	OilRig	+.62	+.62	[+0.16, +1.06]	above	above		
R-F	Sandworm	+.31	+.33	[-0.01, +0.68]	spans	spans		
R-F	menuPass	+.91	+.91	[+0.33, +1.49]	above	above		
R-F	Attr.	3/4	3/4					
R-G	APT29	+.30	+.30	[+0.05, +0.55]	above	above		
R-G	OilRig	+.55	+.55	[-0.01, +1.08]	spans	spans	*	*
R-G	Sandworm	+.18	+.18	[-0.15, +0.52]	spans	spans		
R-G	menuPass	+.39	+.39	[-0.31, +1.13]	spans	spans	*	*
R-G	Attr.	3/4	3/4					
R-Q	APT29	+.12	+.21	[-0.02, +0.45]	spans	spans		*
R-Q	OilRig	-.17	-.12	[-0.43, +0.22]	spans	spans	*	
R-Q	Sandworm	+.17	+.15	[-0.14, +0.45]	spans	spans		
R-Q	menuPass	+1.00	+.87	[+0.52, +1.09]	above	above		
R-Q	Attr.	1/4	2/4					
R-Q5	APT29	+.10	+.10	[-0.13, +0.33]	spans	spans		
R-Q5	OilRig	-.21	-.17	[-0.38, +0.06]	spans	spans	*	
R-Q5	Sandworm	-.02	+.02	[-0.12, +0.17]	spans	spans		
R-Q5	menuPass	+.86	+.82	[+0.30, +1.31]	above	above		
R-Q5	Attr.	1/4	2/4					
R-M	APT29	+.16	+.14	[-0.08, +0.38]	spans	spans		
R-M	OilRig	+.37	+.20	[-0.14, +0.57]	above	spans		
R-M	Sandworm	-.04	-.02	[-0.22, +0.23]	spans	spans		
R-M	menuPass	+1.04	+.93	[+0.30, +1.54]	above	above		
R-M	Attr.	3/4	3/4					
R-M5	APT29	+.13	+.09	[-0.12, +0.31]	spans	spans		
R-M5	OilRig	+.07	+.07	[-0.22, +0.39]	spans	spans		
R-M5	Sandworm	+.06	+.01	[-0.15, +0.21]	spans	spans		
R-M5	menuPass	+1.07	+.68	[+0.18, +1.23]	above	above		
R-M5	Attr.	2/4	2/4					
R-M3	APT29	-.24	-.11	[-0.26, +0.06]	below	spans		
R-M3	OilRig	+.08	+.01	[-0.25, +0.30]	spans	spans		
R-M3	Sandworm	+.01	+.01	[-0.07, +0.15]	spans	spans		
R-M3	menuPass	+.89	+.43	[+0.00, +0.89]	above	spans		*
R-M3	Attr.	0/4	1/4					

Table S7: Strict whole-span evidence filter: R−B gaps for every model (cf. main Table 5).

Model	Actor	ΔF1 adopted	ΔF1	boot	jackknife	clear adopted	clear	Δenr adopted	Δenr	boot	jackknife	clear adopted	clear	ΔIG adopted	ΔIG	i.i.d.	status adopted	status
Claude Opus	APT29	+0.137	+0.126	[-0.101, +0.168]	[-0.085, +0.337]	not clear	not clear	+0.03	-0.01	[-0.86, +0.46]	[-0.97, +0.95]	not clear	not clear	+0.06	+0.02	[-0.18, +0.24]	spans	spans
Claude Opus	OilRig	+0.280	+0.275	[+0.182, +0.325]	[+0.163, +0.387]	above	above	+1.86	+1.82	[+1.26, +2.98]	[+0.21, +3.43]	above	above	+0.61	+0.61	[+0.17, +1.03]	above	above
Claude Opus	Sandworm	+0.189	+0.203	[-0.009, +0.264]	[-0.014, +0.420]	not clear	not clear	+0.80	+0.97	[+0.19, +2.15]	[-0.28, +2.22]	not clear	not clear	+0.33	+0.35	[-0.04, +0.72]	spans	spans
Claude Opus	menuPass	-0.195	-0.189	[-0.491, -0.057]	[-0.538, +0.160]	not clear	not clear	-5.08	-4.93	[-10.73, -1.31]	[-11.79, +1.94]	not clear	not clear	-0.09	-0.09	[-0.51, +0.26]	spans	spans
GPT-6 Astra	APT29	+0.273	+0.273	[+0.023, +0.342]	[+0.021, +0.526]	above	above	+0.56	+0.56	[-0.41, +1.33]	[-0.51, +1.64]	not clear	not clear	+0.30	+0.30	[+0.06, +0.53]	above	above
GPT-6 Astra	OilRig	+0.193	+0.193	[+0.115, +0.297]	[+0.098, +0.288]	above	above	+0.34	+0.34	[-0.56, +2.16]	[-0.80, +1.49]	not clear	not clear	+0.50	+0.50	[-0.02, +1.00]	spans	spans
GPT-6 Astra	Sandworm	+0.150	+0.150	[-0.092, +0.217]	[-0.280, +0.580]	not clear	not clear	-0.22	-0.22	[-1.69, +0.64]	[-2.73, +2.28]	not clear	not clear	+0.17	+0.17	[-0.14, +0.46]	spans	spans
GPT-6 Astra	menuPass	-0.143	-0.143	[-0.307, -0.082]	[-0.223, -0.062]	below	below	-5.57	-5.57	[-8.85, -3.47]	[-6.87, -4.26]	below	below	-0.01	-0.01	[-0.92, +0.74]	spans	spans
Qwen 14B	APT29	+0.158	+0.187	[-0.042, +0.221]	[-0.194, +0.568]	not clear	not clear	-0.17	+0.13	[-0.46, +0.62]	[-1.39, +1.65]	not clear	not clear	+0.02	+0.11	[-0.04, +0.27]	spans	spans
Qwen 14B	OilRig	+0.134	+0.151	[+0.003, +0.187]	[-0.068, +0.369]	not clear	not clear	+1.09	+1.38	[+0.08, +2.10]	[-0.60, +3.35]	not clear	not clear	-0.06	-0.01	[-0.38, +0.37]	spans	spans
Qwen 14B	Sandworm	+0.222	+0.209	[-0.044, +0.216]	[-0.113, +0.531]	not clear	not clear	+1.10	+1.14	[+0.11, +1.30]	[-0.04, +2.32]	above	not clear	+0.25	+0.22	[-0.18, +0.60]	spans	spans
Qwen 14B	menuPass	+0.060	+0.102	[-0.211, +0.102]	[-0.338, +0.542]	not clear	not clear	+0.23	+1.17	[-4.67, +1.17]	[-6.15, +8.48]	not clear	not clear	+0.68	+0.55	[-0.05, +1.10]	above	spans
Qwen 14B sampled	APT29	+0.160	+0.161	[-0.018, +0.202]	[-0.010, +0.333]	not clear	not clear	-0.15	-0.11	[-0.90, +0.72]	[-1.02, +0.81]	not clear	not clear	+0.00	+0.00	[-0.16, +0.17]	spans	spans
Qwen 14B sampled	OilRig	+0.154	+0.153	[-0.048, +0.248]	[-0.210, +0.517]	not clear	not clear	+1.23	+1.32	[-0.71, +2.32]	[-2.18, +4.83]	not clear	not clear	-0.10	-0.06	[-0.28, +0.18]	spans	spans
Qwen 14B sampled	Sandworm	+0.130	+0.129	[-0.031, +0.178]	[+0.045, +0.213]	not clear	not clear	+0.49	+0.71	[-0.40, +1.33]	[+0.00, +1.43]	not clear	not clear	+0.06	+0.09	[-0.05, +0.25]	spans	spans
Qwen 14B sampled	menuPass	+0.019	+0.177	[-0.220, +0.267]	[-0.345, +0.700]	not clear	not clear	-0.54	+3.89	[-4.98, +8.76]	[-8.82, +16.60]	not clear	not clear	+0.51	+0.47	[+0.04, +0.93]	above	above
Ministral 14B	APT29	+0.233	+0.220	[-0.005, +0.264]	[-0.042, +0.482]	not clear	not clear	+0.95	+1.28	[-0.15, +2.17]	[-0.78, +3.34]	not clear	not clear	+0.17	+0.16	[-0.06, +0.39]	spans	spans
Ministral 14B	OilRig	+0.375	+0.350	[+0.148, +0.400]	[-0.015, +0.715]	above	not clear	+3.31	+3.40	[+2.02, +4.28]	[+0.48, +6.32]	above	above	+0.56	+0.39	[+0.03, +0.78]	above	above
Ministral 14B	Sandworm	+0.049	+0.054	[-0.118, +0.064]	[-0.057, +0.164]	not clear	not clear	+0.40	+0.52	[-1.33, +3.00]	[-1.88, +2.93]	not clear	not clear	-0.03	-0.01	[-0.34, +0.32]	spans	spans
Ministral 14B	menuPass	+0.044	+0.085	[-0.068, +0.159]	[-0.068, +0.239]	not clear	not clear	+0.00	+1.01	[-0.72, +3.61]	[-1.00, +3.02]	not clear	not clear	+0.72	+0.61	[-0.38, +1.47]	spans	spans
Ministral 14B sampled	APT29	+0.143	+0.122	[-0.076, +0.188]	[-0.060, +0.304]	not clear	not clear	-0.08	-0.07	[-1.05, +0.68]	[-0.93, +0.79]	not clear	not clear	+0.12	+0.08	[-0.08, +0.24]	spans	spans
Ministral 14B sampled	OilRig	+0.234	+0.232	[+0.134, +0.267]	[+0.207, +0.258]	above	above	+1.99	+2.28	[+1.53, +3.50]	[+1.42, +3.13]	above	above	+0.26	+0.26	[-0.02, +0.60]	spans	spans
Ministral 14B sampled	Sandworm	+0.161	+0.104	[-0.104, +0.154]	[-0.117, +0.326]	not clear	not clear	+1.04	+0.91	[-1.30, +1.99]	[-0.21, +2.04]	not clear	not clear	+0.07	+0.02	[-0.26, +0.30]	spans	spans
Ministral 14B sampled	menuPass	+0.025	+0.022	[-0.223, +0.109]	[-0.159, +0.202]	not clear	not clear	-0.54	+0.07	[-4.34, +2.25]	[-1.09, +1.22]	not clear	not clear	+0.64	+0.26	[-0.28, +0.99]	above	spans
Ministral 3B	APT29	+0.133	+0.130	[+0.000, +0.143]	[-0.047, +0.308]	not clear	not clear	–	–	–	–	undefined	undefined	-0.24	-0.11	[-0.26, +0.06]	below	spans
Ministral 3B	OilRig	+0.204	+0.133	[+0.000, +0.160]	[-0.061, +0.328]	above	not clear	–	–	–	–	undefined	undefined	+0.08	+0.01	[-0.25, +0.30]	spans	spans
Ministral 3B	Sandworm	+0.162	+0.077	[+0.000, +0.083]	[-0.092, +0.246]	above	not clear	–	–	–	–	undefined	undefined	+0.01	+0.01	[-0.07, +0.15]	spans	spans
Ministral 3B	menuPass	+0.229	+0.222	[+0.000, +0.308]	[-0.113, +0.557]	not clear	not clear	–	–	–	–	undefined	undefined	+0.89	+0.43	[+0.00, +0.89]	above	spans

Table S8: Each local model's edge-F1 as a share of Claude Opus's (the 80% criterion of the small-model comparison): as run, and under each evidence filter against Claude Opus filtered the same way or as run.

Model	Actor	as run	fragment vs Opus filtered	fragment vs Opus as run	strict vs Opus filtered	strict vs Opus as run
Qwen 14B	APT29	96%	104%	104%	107%	104%
Qwen 14B	OilRig	52%	56%	56%	57%	56%
Qwen 14B	Sandworm	91%	88%	88%	85%	88%
Qwen 14B	menuPass	90%	101%	101%	102%	104%
Qwen 14B	meets 80%	3/4	3/4	3/4	3/4	3/4
Qwen 14B sampled	APT29	97%	96%	96%	100%	97%
Qwen 14B sampled	OilRig	50%	51%	51%	51%	50%
Qwen 14B sampled	Sandworm	63%	66%	66%	61%	63%
Qwen 14B sampled	menuPass	80%	125%	125%	130%	133%
Qwen 14B sampled	meets 80%	1/4	2/4	2/4	2/4	2/4
Ministral 14B	APT29	90%	86%	86%	89%	86%
Ministral 14B	OilRig	95%	90%	90%	90%	88%
Ministral 14B	Sandworm	45%	46%	46%	44%	46%
Ministral 14B	menuPass	89%	103%	103%	101%	103%
Ministral 14B	meets 80%	3/4	3/4	3/4	3/4	3/4
Ministral 14B sampled	APT29	84%	78%	78%	81%	78%
Ministral 14B sampled	OilRig	63%	59%	59%	63%	63%
Ministral 14B sampled	Sandworm	71%	56%	56%	54%	56%
Ministral 14B sampled	menuPass	97%	94%	94%	94%	96%
Ministral 14B sampled	meets 80%	2/4	1/4	1/4	2/4	1/4
Ministral 3B	APT29	36%	35%	35%	36%	35%
Ministral 3B	OilRig	52%	34%	34%	34%	34%
Ministral 3B	Sandworm	44%	19%	19%	20%	21%
Ministral 3B	menuPass	76%	100%	100%	73%	74%
Ministral 3B	meets 80%	0/4	1/4	1/4	0/4	0/4
GPT-6 Astra	APT29	121%	121%	121%	125%	121%
GPT-6 Astra	OilRig	86%	86%	86%	87%	86%
GPT-6 Astra	Sandworm	89%	89%	89%	86%	89%
GPT-6 Astra	menuPass	60%	60%	60%	59%	60%
GPT-6 Astra	meets 80%	3/4	3/4	3/4	3/4	3/4

2 Ground-truth variants

Two alternative readings of the plans, each scored with the same kernels. Variant B treats menuPass's two steps marked Optional (8.A and 7.1) as separate segments, which leaves the plan four edges. Variant A splits OilRig and Sandworm at every Linux/Windows handoff (OilRig 15 → 12 edges, Sandworm 21 → 19). Neither is adopted.

Table S9: Variant B (menuPass optional steps as separate segments; four edges): support recovery, adopted vs. variant.

Kernel	Actor	GT edges	F1 adopted	F1	F1/chance adopted	F1/chance	pct adopted	pct
R-F	APT29	28	0.371	0.371	+1.99	+1.99	99.8	99.8
R-F	OilRig	15	0.396	0.396	+3.23	+3.23	100.0	100.0
R-F	Sandworm	21	0.372	0.372	+2.65	+2.65	99.9	99.9
R-F	menuPass	4	0.300	0.308	+5.78	+8.28	99.9	100.0
R-G	APT29	28	0.450	0.450	+2.21	+2.21	100.0	100.0
R-G	OilRig	15	0.341	0.341	+2.72	+2.72	100.0	100.0
R-G	Sandworm	21	0.331	0.331	+2.22	+2.22	99.7	99.7
R-G	menuPass	4	0.181	0.144	+3.30	+3.73	97.5	95.6
R-Q	APT29	28	0.358	0.358	+2.00	+2.00	99.7	99.7
R-Q	OilRig	15	0.205	0.205	+2.02	+2.02	92.6	92.6
R-Q	Sandworm	21	0.340	0.340	+2.44	+2.44	99.8	99.8
R-Q	menuPass	4	0.270	0.229	+4.89	+5.87	100.0	100.0
R-Q5	APT29	28	0.359	0.359	+1.98	+1.98	97.5	97.5
R-Q5	OilRig	15	0.199	0.199	+1.88	+1.88	83.0	83.0
R-Q5	Sandworm	21	0.236	0.236	+1.71	+1.71	86.6	86.6
R-Q5	menuPass	4	0.239	0.193	+4.44	+5.06	99.5	99.2
R-M	APT29	28	0.333	0.333	+2.03	+2.03	99.4	99.4
R-M	OilRig	15	0.375	0.375	+3.31	+3.31	100.0	100.0
R-M	Sandworm	21	0.167	0.167	+1.73	+1.73	84.1	84.1
R-M	menuPass	4	0.267	0.214	+5.06	+5.69	99.8	99.6
R-M5	APT29	28	0.311	0.311	+1.71	+1.71	96.0	96.0
R-M5	OilRig	15	0.250	0.250	+2.21	+2.21	97.0	97.0
R-M5	Sandworm	21	0.265	0.265	+2.48	+2.48	96.0	96.0
R-M5	menuPass	4	0.290	0.240	+5.50	+6.38	99.6	99.0
R-M3	APT29	28	0.133	0.133	+0.81	+0.81	32.6	32.6
R-M3	OilRig	15	0.204	0.204	+1.78	+1.78	91.9	91.9
R-M3	Sandworm	21	0.162	0.162	+1.62	+1.62	81.5	81.5
R-M3	menuPass	4	0.229	0.182	+4.18	+4.71	99.7	99.3
B-F	APT29	28	0.234	0.234	+1.96	+1.96	94.3	94.3
B-F	OilRig	15	0.116	0.116	+1.37	+1.37	66.9	66.9
B-F	Sandworm	21	0.183	0.183	+1.85	+1.85	81.9	81.9
B-F	menuPass	4	0.495	0.460	+10.87	+13.60	100.0	100.0
B-G	APT29	28	0.177	0.177	+1.65	+1.65	75.9	75.9
B-G	OilRig	15	0.149	0.149	+2.37	+2.37	78.5	78.5
B-G	Sandworm	21	0.181	0.181	+2.45	+2.45	91.3	91.3
B-G	menuPass	4	0.323	0.380	+8.87	+13.30	99.0	99.6
B-Q	APT29	28	0.200	0.200	+2.17	+2.17	94.3	94.3
B-Q	OilRig	15	0.071	0.071	+0.93	+0.93	51.1	51.1
B-Q	Sandworm	21	0.118	0.118	+1.33	+1.33	68.3	68.3
B-Q	menuPass	4	0.211	0.118	+4.67	+3.50	96.7	85.8
B-M	APT29	28	0.100	0.100	+1.08	+1.08	57.3	57.3
B-M	OilRig	15	0.000	0.000	–	–	17.2	17.2
B-M	Sandworm	21	0.118	0.118	+1.33	+1.33	68.3	68.3
B-M	menuPass	4	0.222	0.125	+5.06	+3.79	97.2	86.9
ATTK	APT29	28	0.238	0.238	+0.94	+0.94	27.4	27.4
ATTK	OilRig	15	0.204	0.204	+1.38	+1.38	99.7	99.7
ATTK	Sandworm	21	0.207	0.207	+1.00	+1.00	50.0	50.0
ATTK	menuPass	4	0.069	0.053	+1.10	+1.24	60.5	69.4

Table S10: Variant B: information gain over uniform, adopted vs. variant.

Kernel	Plan	IG adopted	IG	i.i.d. 95%	status adopted	status	* adopted	*
R-F	APT29	+.11	+.11	[-0.13, +0.37]	spans	spans		
R-F	OilRig	+.62	+.62	[+0.17, +1.04]	above	above		
R-F	Sandworm	+.31	+.31	[-0.05, +0.68]	spans	spans		
R-F	menuPass	+.91	+1.40	[+1.15, +1.74]	above	above		
R-F	Attr.	3/4	3/4					
R-G	APT29	+.30	+.30	[+0.05, +0.55]	above	above		
R-G	OilRig	+.55	+.55	[-0.01, +1.08]	spans	spans	*	*
R-G	Sandworm	+.18	+.18	[-0.15, +0.52]	spans	spans		
R-G	menuPass	+.39	+.49	[-0.59, +1.58]	spans	spans	*	*
R-G	Attr.	3/4	3/4					
R-Q	APT29	+.12	+.12	[-0.12, +0.37]	spans	spans		
R-Q	OilRig	-.17	-.17	[-0.45, +0.14]	spans	spans	*	*
R-Q	Sandworm	+.17	+.17	[-0.11, +0.46]	spans	spans		
R-Q	menuPass	+1.00	+1.12	[+0.87, +1.38]	above	above		
R-Q	Attr.	1/4	2/4					
R-Q5	APT29	+.10	+.10	[-0.15, +0.35]	spans	spans		
R-Q5	OilRig	-.21	-.21	[-0.42, +0.03]	spans	spans	*	*
R-Q5	Sandworm	-.02	-.02	[-0.18, +0.17]	spans	spans		
R-Q5	menuPass	+.86	+.97	[+0.24, +1.47]	above	above		
R-Q5	Attr.	1/4	1/4					
R-M	APT29	+.16	+.16	[-0.10, +0.42]	spans	spans		
R-M	OilRig	+.37	+.37	[+0.07, +0.69]	above	above		
R-M	Sandworm	-.04	-.04	[-0.26, +0.24]	spans	spans		
R-M	menuPass	+1.04	+1.07	[+0.22, +1.72]	above	above		
R-M	Attr.	3/4	3/4					
R-M5	APT29	+.13	+.13	[-0.10, +0.38]	spans	spans		
R-M5	OilRig	+.07	+.07	[-0.28, +0.42]	spans	spans		
R-M5	Sandworm	+.06	+.06	[-0.17, +0.31]	spans	spans		
R-M5	menuPass	+1.07	+1.15	[+0.71, +1.59]	above	above		
R-M5	Attr.	2/4	2/4					
R-M3	APT29	-.24	-.24	[-0.41, -0.04]	below	below		
R-M3	OilRig	+.08	+.08	[-0.32, +0.51]	spans	spans		
R-M3	Sandworm	+.01	+.01	[-0.15, +0.21]	spans	spans		
R-M3	menuPass	+.89	+.89	[+0.14, +1.33]	above	above		
R-M3	Attr.	0/4	0/4					
B-F	APT29	+.06	+.06	[-0.08, +0.22]	spans	spans		
B-F	OilRig	+.01	+.01	[-0.18, +0.25]	spans	spans		
B-F	Sandworm	-.02	-.02	[-0.17, +0.16]	spans	spans		
B-F	menuPass	+1.01	+1.30	[+1.22, +1.38]	above	above		
B-F	Attr.	2/4	2/4					
B-G	APT29	+.00	+.00	[-0.13, +0.16]	spans	spans		
B-G	OilRig	+.05	+.05	[-0.07, +0.21]	spans	spans		
B-G	Sandworm	+.01	+.01	[-0.14, +0.20]	spans	spans		
B-G	menuPass	+.40	+.64	[-0.10, +1.38]	spans	spans	*	*
B-G	Attr.	2/4	2/4					
B-Q	APT29	+.10	+.10	[-0.06, +0.29]	spans	spans		
B-Q	OilRig	-.11	-.11	[-0.21, +0.07]	spans	spans		
B-Q	Sandworm	-.07	-.07	[-0.21, +0.12]	spans	spans		
B-Q	menuPass	+.32	+.19	[-0.21, +0.98]	spans	spans	*	*
B-Q	Attr.	1/4	1/4					
B-M	APT29	-.01	-.01	[-0.11, +0.11]	spans	spans		
B-M	OilRig	-.19	-.19	[-0.26, -0.12]	below	below		
B-M	Sandworm	-.01	-.01	[-0.20, +0.19]	spans	spans		
B-M	menuPass	+.32	+.19	[-0.21, +0.98]	spans	spans	*	*
B-M	Attr.	1/4	1/4					
ATTK	APT29	-.61	-.61	[-1.27, -0.02]	below	below	*	*
ATTK	OilRig	-.05	-.05	[-0.72, +0.55]	spans	spans		
ATTK	Sandworm	-.30	-.30	[-0.59, -0.06]	below	below		
ATTK	menuPass	-.92	-1.30	[-4.39, +0.63]	spans	spans	*	*
ATTK	Attr.	1/4	1/4					

Table S11: Variant B: R−B gaps for every model, adopted vs. variant.

Model	Actor	ΔF1 adopted	ΔF1	boot	jackknife	clear adopted	clear	Δenr adopted	Δenr	boot	jackknife	clear adopted	clear	ΔIG adopted	ΔIG	i.i.d.	status adopted	status
Claude Opus	APT29	+0.137	+0.137	[-0.098, +0.177]	[-0.065, +0.339]	not clear	not clear	+0.03	+0.03	[-0.85, +0.44]	[-0.87, +0.93]	not clear	not clear	+0.06	+0.06	[-0.15, +0.27]	spans	spans
Claude Opus	OilRig	+0.280	+0.280	[+0.185, +0.329]	[+0.181, +0.380]	above	above	+1.86	+1.86	[+1.29, +2.98]	[+0.42, +3.30]	above	above	+0.61	+0.61	[+0.18, +1.02]	above	above
Claude Opus	Sandworm	+0.189	+0.189	[-0.012, +0.255]	[-0.009, +0.386]	not clear	not clear	+0.80	+0.80	[-0.05, +1.79]	[-0.30, +1.90]	not clear	not clear	+0.33	+0.33	[-0.07, +0.72]	spans	spans
Claude Opus	menuPass	-0.195	-0.152	[-0.465, -0.004]	[-0.536, +0.232]	not clear	not clear	-5.08	-5.32	[-13.80, -0.15]	[-15.80, +5.16]	not clear	not clear	-0.09	+0.11	[-0.17, +0.39]	spans	spans
GPT-6 Astra	APT29	+0.273	+0.273	[+0.023, +0.342]	[+0.021, +0.526]	above	above	+0.56	+0.56	[-0.41, +1.33]	[-0.51, +1.64]	not clear	not clear	+0.30	+0.30	[+0.06, +0.53]	above	above
GPT-6 Astra	OilRig	+0.193	+0.193	[+0.115, +0.297]	[+0.098, +0.288]	above	above	+0.34	+0.34	[-0.56, +2.16]	[-0.80, +1.49]	not clear	not clear	+0.50	+0.50	[-0.02, +1.00]	spans	spans
GPT-6 Astra	Sandworm	+0.150	+0.150	[-0.092, +0.217]	[-0.280, +0.580]	not clear	not clear	-0.22	-0.22	[-1.69, +0.64]	[-2.73, +2.28]	not clear	not clear	+0.17	+0.17	[-0.14, +0.46]	spans	spans
GPT-6 Astra	menuPass	-0.143	-0.235	[-0.375, -0.136]	[-0.369, -0.102]	below	below	-5.57	-9.58	[-13.79, -5.91]	[-13.10, -6.05]	below	below	-0.01	-0.14	[-1.45, +0.95]	spans	spans
Qwen 14B	APT29	+0.158	+0.158	[-0.067, +0.207]	[-0.213, +0.530]	not clear	not clear	-0.17	-0.17	[-1.02, +0.43]	[-1.83, +1.50]	not clear	not clear	+0.02	+0.02	[-0.14, +0.19]	spans	spans
Qwen 14B	OilRig	+0.134	+0.134	[+0.000, +0.171]	[-0.079, +0.346]	not clear	not clear	+1.09	+1.09	[+0.00, +1.76]	[-0.92, +3.10]	not clear	not clear	-0.06	-0.06	[-0.41, +0.29]	spans	spans
Qwen 14B	Sandworm	+0.222	+0.222	[-0.044, +0.229]	[-0.053, +0.497]	not clear	not clear	+1.10	+1.10	[+0.11, +1.30]	[+0.05, +2.16]	above	above	+0.25	+0.25	[-0.11, +0.59]	spans	spans
Qwen 14B	menuPass	+0.060	+0.111	[-0.118, +0.190]	[-0.237, +0.459]	not clear	not clear	+0.23	+2.37	[-3.50, +4.77]	[-6.35, +11.10]	not clear	not clear	+0.68	+0.93	[+0.40, +1.32]	above	above
Qwen 14B sampled	APT29	+0.160	+0.160	[-0.019, +0.200]	[-0.013, +0.334]	not clear	not clear	-0.15	-0.15	[-0.94, +0.67]	[-1.15, +0.85]	not clear	not clear	+0.00	+0.00	[-0.17, +0.18]	spans	spans
Qwen 14B sampled	OilRig	+0.154	+0.154	[-0.034, +0.244]	[-0.144, +0.452]	not clear	not clear	+1.23	+1.23	[-0.52, +2.18]	[-1.52, +3.98]	not clear	not clear	-0.10	-0.10	[-0.33, +0.15]	spans	spans
Qwen 14B sampled	Sandworm	+0.130	+0.130	[-0.016, +0.174]	[+0.076, +0.183]	not clear	not clear	+0.49	+0.49	[-0.32, +1.11]	[-0.30, +1.29]	not clear	not clear	+0.06	+0.06	[-0.08, +0.22]	spans	spans
Qwen 14B sampled	menuPass	+0.019	+0.070	[-0.125, +0.321]	[-0.283, +0.423]	not clear	not clear	-0.54	+1.33	[-3.79, +10.77]	[-8.69, +11.34]	not clear	not clear	+0.51	+0.78	[+0.31, +1.25]	above	above
Ministral 14B	APT29	+0.233	+0.233	[-0.009, +0.270]	[+0.019, +0.448]	not clear	not clear	+0.95	+0.95	[-0.27, +1.52]	[-0.07, +1.97]	not clear	not clear	+0.17	+0.17	[-0.04, +0.40]	spans	spans
Ministral 14B	OilRig	+0.375	+0.375	[+0.222, +0.400]	[+0.147, +0.603]	above	above	+3.31	+3.31	[+2.85, +3.88]	[+2.05, +4.57]	above	above	+0.56	+0.56	[+0.22, +0.91]	above	above
Ministral 14B	Sandworm	+0.049	+0.049	[-0.044, +0.059]	[-0.022, +0.120]	not clear	not clear	+0.40	+0.40	[+0.11, +3.00]	[-0.54, +1.34]	not clear	not clear	-0.03	-0.03	[-0.38, +0.34]	spans	spans
Ministral 14B	menuPass	+0.044	+0.089	[-0.125, +0.183]	[+0.011, +0.167]	not clear	not clear	+0.00	+1.90	[-3.79, +6.32]	[-0.14, +3.93]	not clear	not clear	+0.72	+0.88	[-0.76, +1.92]	spans	spans
Ministral 14B sampled	APT29	+0.143	+0.143	[-0.069, +0.197]	[-0.007, +0.293]	not clear	not clear	-0.08	-0.08	[-1.07, +0.57]	[-0.66, +0.51]	not clear	not clear	+0.12	+0.12	[-0.07, +0.32]	spans	spans
Ministral 14B sampled	OilRig	+0.234	+0.234	[+0.157, +0.274]	[+0.158, +0.311]	above	above	+1.99	+1.99	[+1.40, +3.09]	[+0.74, +3.24]	above	above	+0.26	+0.26	[-0.08, +0.64]	spans	spans
Ministral 14B sampled	Sandworm	+0.161	+0.161	[-0.046, +0.229]	[-0.131, +0.453]	not clear	not clear	+1.04	+1.04	[-0.15, +2.08]	[-0.80, +2.88]	not clear	not clear	+0.07	+0.07	[-0.25, +0.39]	spans	spans
Ministral 14B sampled	menuPass	+0.025	+0.066	[-0.192, +0.185]	[-0.058, +0.190]	not clear	not clear	-0.54	+1.09	[-5.75, +5.83]	[-2.21, +4.39]	not clear	not clear	+0.64	+0.83	[+0.00, +1.67]	above	above
Ministral 3B	APT29	+0.133	+0.133	[+0.000, +0.157]	[+0.001, +0.266]	not clear	not clear	–	–	–	–	undefined	undefined	-0.24	-0.24	[-0.41, -0.04]	below	below
Ministral 3B	OilRig	+0.204	+0.204	[+0.059, +0.256]	[+0.016, +0.392]	above	above	–	–	–	–	undefined	undefined	+0.08	+0.08	[-0.32, +0.51]	spans	spans
Ministral 3B	Sandworm	+0.162	+0.162	[+0.062, +0.187]	[+0.008, +0.317]	above	above	–	–	–	–	undefined	undefined	+0.01	+0.01	[-0.15, +0.21]	spans	spans
Ministral 3B	menuPass	+0.229	+0.182	[+0.000, +0.300]	[-0.113, +0.477]	not clear	not clear	–	–	–	–	undefined	undefined	+0.89	+0.89	[+0.14, +1.33]	above	above

Table S12: Variant A (split at every Linux/Windows handoff): support recovery, adopted vs. variant.

Kernel	Actor	GT edges	F1 adopted	F1	F1/chance adopted	F1/chance	pct adopted	pct
R-F	APT29	28	0.371	0.371	+1.99	+1.99	99.8	99.8
R-F	OilRig	12	0.396	0.310	+3.23	+3.00	100.0	99.9
R-F	Sandworm	19	0.372	0.348	+2.65	+2.63	99.9	99.9
R-F	menuPass	6	0.300	0.300	+5.78	+5.78	99.9	99.9
R-G	APT29	28	0.450	0.450	+2.21	+2.21	100.0	100.0
R-G	OilRig	12	0.341	0.259	+2.72	+2.45	100.0	99.6
R-G	Sandworm	19	0.331	0.308	+2.22	+2.21	99.7	99.4
R-G	menuPass	6	0.181	0.181	+3.30	+3.30	97.5	97.5
R-Q	APT29	28	0.358	0.358	+2.00	+2.00	99.7	99.7
R-Q	OilRig	12	0.205	0.222	+2.02	+2.53	92.6	96.6
R-Q	Sandworm	19	0.340	0.353	+2.44	+2.69	99.8	99.9
R-Q	menuPass	6	0.270	0.270	+4.89	+4.89	100.0	100.0
R-Q5	APT29	28	0.359	0.359	+1.98	+1.98	97.5	97.5
R-Q5	OilRig	12	0.199	0.190	+1.88	+2.10	83.0	89.5
R-Q5	Sandworm	19	0.236	0.241	+1.71	+1.86	86.6	89.6
R-Q5	menuPass	6	0.239	0.239	+4.44	+4.44	99.5	99.5
R-M	APT29	28	0.333	0.333	+2.03	+2.03	99.4	99.4
R-M	OilRig	12	0.375	0.356	+3.31	+3.68	100.0	100.0
R-M	Sandworm	19	0.167	0.176	+1.73	+1.92	84.1	87.5
R-M	menuPass	6	0.267	0.267	+5.06	+5.06	99.8	99.8
R-M5	APT29	28	0.311	0.311	+1.71	+1.71	96.0	96.0
R-M5	OilRig	12	0.250	0.259	+2.21	+2.69	97.0	98.8
R-M5	Sandworm	19	0.265	0.279	+2.48	+2.74	96.0	97.2
R-M5	menuPass	6	0.290	0.290	+5.50	+5.50	99.6	99.6
R-M3	APT29	28	0.133	0.133	+0.81	+0.81	32.6	32.6
R-M3	OilRig	12	0.204	0.217	+1.78	+2.23	91.9	96.9
R-M3	Sandworm	19	0.162	0.171	+1.62	+1.80	81.5	85.4
R-M3	menuPass	6	0.229	0.229	+4.18	+4.18	99.7	99.7
B-F	APT29	28	0.234	0.234	+1.96	+1.96	94.3	94.3
B-F	OilRig	12	0.116	0.115	+1.37	+1.55	66.9	71.3
B-F	Sandworm	19	0.183	0.194	+1.85	+2.04	81.9	84.7
B-F	menuPass	6	0.495	0.495	+10.87	+10.87	100.0	100.0
B-G	APT29	28	0.177	0.177	+1.65	+1.65	75.9	75.9
B-G	OilRig	12	0.149	0.170	+2.37	+2.97	78.5	81.5
B-G	Sandworm	19	0.181	0.193	+2.45	+2.71	91.3	93.0
B-G	menuPass	6	0.323	0.323	+8.87	+8.87	99.0	99.0
B-Q	APT29	28	0.200	0.200	+2.17	+2.17	94.3	94.3
B-Q	OilRig	12	0.071	0.000	+0.93	–	51.1	20.0
B-Q	Sandworm	19	0.118	0.125	+1.33	+1.47	68.3	72.8
B-Q	menuPass	6	0.211	0.211	+4.67	+4.67	96.7	96.7
B-M	APT29	28	0.100	0.100	+1.08	+1.08	57.3	57.3
B-M	OilRig	12	0.000	0.000	–	–	17.2	21.5
B-M	Sandworm	19	0.118	0.125	+1.33	+1.47	68.3	72.8
B-M	menuPass	6	0.222	0.222	+5.06	+5.06	97.2	97.2
ATTK	APT29	28	0.238	0.238	+0.94	+0.94	27.4	27.4
ATTK	OilRig	12	0.204	0.167	+1.38	+1.38	99.7	99.1
ATTK	Sandworm	19	0.207	0.189	+1.00	+1.00	50.0	50.0
ATTK	menuPass	6	0.069	0.069	+1.10	+1.10	60.5	60.5

Table S13: Variant A: information gain over uniform, adopted vs. variant.

Kernel	Plan	IG adopted	IG	i.i.d. 95%	status adopted	status	* adopted	*
R-F	APT29	+.11	+.11	[-0.13, +0.37]	spans	spans		
R-F	OilRig	+.62	+.61	[+0.07, +1.12]	above	above		*
R-F	Sandworm	+.31	+.31	[-0.09, +0.72]	spans	spans		
R-F	menuPass	+.91	+.91	[+0.33, +1.49]	above	above		
R-F	Attr.	3/4	3/4					
R-G	APT29	+.30	+.30	[+0.05, +0.55]	above	above		
R-G	OilRig	+.55	+.53	[-0.14, +1.17]	spans	spans	*	*
R-G	Sandworm	+.18	+.16	[-0.20, +0.54]	spans	spans		
R-G	menuPass	+.39	+.39	[-0.31, +1.13]	spans	spans	*	*
R-G	Attr.	3/4	3/4					
R-Q	APT29	+.12	+.12	[-0.12, +0.37]	spans	spans		
R-Q	OilRig	-.17	-.13	[-0.46, +0.24]	spans	spans	*	*
R-Q	Sandworm	+.17	+.25	[-0.05, +0.54]	spans	spans		
R-Q	menuPass	+1.00	+1.00	[+0.55, +1.37]	above	above		
R-Q	Attr.	1/4	1/4					
R-Q5	APT29	+.10	+.10	[-0.15, +0.35]	spans	spans		
R-Q5	OilRig	-.21	-.20	[-0.45, +0.07]	spans	spans	*	*
R-Q5	Sandworm	-.02	+.00	[-0.18, +0.21]	spans	spans		
R-Q5	menuPass	+.86	+.86	[+0.35, +1.36]	above	above		
R-Q5	Attr.	1/4	2/4					
R-M	APT29	+.16	+.16	[-0.10, +0.42]	spans	spans		
R-M	OilRig	+.37	+.45	[+0.08, +0.81]	above	above		
R-M	Sandworm	-.04	+.01	[-0.23, +0.32]	spans	spans		
R-M	menuPass	+1.04	+1.04	[+0.35, +1.73]	above	above		
R-M	Attr.	3/4	3/4					
R-M5	APT29	+.13	+.13	[-0.10, +0.38]	spans	spans		
R-M5	OilRig	+.07	+.18	[-0.20, +0.57]	spans	spans		
R-M5	Sandworm	+.06	+.11	[-0.13, +0.38]	spans	spans		
R-M5	menuPass	+1.07	+1.07	[+0.54, +1.57]	above	above		
R-M5	Attr.	2/4	3/4					
R-M3	APT29	-.24	-.24	[-0.41, -0.04]	below	below		
R-M3	OilRig	+.08	+.15	[-0.31, +0.64]	spans	spans		
R-M3	Sandworm	+.01	+.04	[-0.14, +0.25]	spans	spans		
R-M3	menuPass	+.89	+.89	[+0.30, +1.42]	above	above		
R-M3	Attr.	0/4	1/4					
B-F	APT29	+.06	+.06	[-0.08, +0.22]	spans	spans		
B-F	OilRig	+.01	+.02	[-0.21, +0.29]	spans	spans		
B-F	Sandworm	-.02	-.00	[-0.17, +0.20]	spans	spans		
B-F	menuPass	+1.01	+1.01	[+0.57, +1.33]	above	above		
B-F	Attr.	2/4	2/4					
B-G	APT29	+.00	+.00	[-0.13, +0.16]	spans	spans		
B-G	OilRig	+.05	+.08	[-0.06, +0.27]	spans	spans		
B-G	Sandworm	+.01	+.02	[-0.14, +0.24]	spans	spans		
B-G	menuPass	+.40	+.40	[-0.12, +0.93]	spans	spans	*	*
B-G	Attr.	2/4	2/4					
B-Q	APT29	+.10	+.10	[-0.06, +0.29]	spans	spans		
B-Q	OilRig	-.11	-.21	[-0.21, -0.21]	spans	below		
B-Q	Sandworm	-.07	-.06	[-0.21, +0.15]	spans	spans		
B-Q	menuPass	+.32	+.32	[-0.21, +0.85]	spans	spans	*	*
B-Q	Attr.	1/4	0/4					
B-M	APT29	-.01	-.01	[-0.11, +0.11]	spans	spans		
B-M	OilRig	-.19	-.20	[-0.27, -0.13]	below	below		
B-M	Sandworm	-.01	+.01	[-0.21, +0.23]	spans	spans		
B-M	menuPass	+.32	+.32	[-0.21, +0.85]	spans	spans	*	*
B-M	Attr.	1/4	1/4					
ATTK	APT29	-.61	-.61	[-1.27, -0.02]	below	below	*	*
ATTK	OilRig	-.05	-.05	[-0.80, +0.60]	spans	spans		*
ATTK	Sandworm	-.30	-.26	[-0.57, -0.01]	below	below		
ATTK	menuPass	-.92	-.92	[-2.99, +0.38]	spans	spans	*	*
ATTK	Attr.	1/4	1/4					

Table S14: Variant A: R−B gaps for every model, adopted vs. variant.

Model	Actor	ΔF1 adopted	ΔF1	boot	jackknife	clear adopted	clear	Δenr adopted	Δenr	boot	jackknife	clear adopted	clear	ΔIG adopted	ΔIG	i.i.d.	status adopted	status
Claude Opus	APT29	+0.137	+0.137	[-0.098, +0.177]	[-0.065, +0.339]	not clear	not clear	+0.03	+0.03	[-0.85, +0.44]	[-0.87, +0.93]	not clear	not clear	+0.06	+0.06	[-0.15, +0.27]	spans	spans
Claude Opus	OilRig	+0.280	+0.195	[+0.123, +0.243]	[+0.141, +0.248]	above	above	+1.86	+1.45	[+1.03, +2.29]	[+1.25, +1.64]	above	above	+0.61	+0.59	[+0.08, +1.08]	above	above
Claude Opus	Sandworm	+0.189	+0.154	[-0.051, +0.226]	[-0.051, +0.359]	not clear	not clear	+0.80	+0.59	[-0.51, +1.71]	[-0.51, +1.70]	not clear	not clear	+0.33	+0.31	[-0.13, +0.74]	spans	spans
Claude Opus	menuPass	-0.195	-0.195	[-0.491, -0.062]	[-0.544, +0.154]	not clear	not clear	-5.08	-5.08	[-10.73, -1.52]	[-11.98, +1.81]	not clear	not clear	-0.09	-0.09	[-0.51, +0.26]	spans	spans
GPT-6 Astra	APT29	+0.273	+0.273	[+0.023, +0.342]	[+0.021, +0.526]	above	above	+0.56	+0.56	[-0.41, +1.33]	[-0.51, +1.64]	not clear	not clear	+0.30	+0.30	[+0.06, +0.53]	above	above
GPT-6 Astra	OilRig	+0.193	+0.089	[+0.013, +0.196]	[+0.038, +0.140]	above	above	+0.34	-0.52	[-1.63, +1.41]	[-0.95, -0.08]	not clear	not clear	+0.50	+0.44	[-0.18, +1.05]	spans	spans
GPT-6 Astra	Sandworm	+0.150	+0.114	[-0.143, +0.194]	[-0.364, +0.592]	not clear	not clear	-0.22	-0.50	[-2.24, +0.53]	[-3.68, +2.68]	not clear	not clear	+0.17	+0.14	[-0.20, +0.47]	spans	spans
GPT-6 Astra	menuPass	-0.143	-0.143	[-0.307, -0.082]	[-0.223, -0.062]	below	below	-5.57	-5.57	[-8.85, -3.47]	[-6.87, -4.26]	below	below	-0.01	-0.01	[-0.92, +0.74]	spans	spans
Qwen 14B	APT29	+0.158	+0.158	[-0.067, +0.207]	[-0.213, +0.530]	not clear	not clear	-0.17	-0.17	[-1.02, +0.43]	[-1.83, +1.50]	not clear	not clear	+0.02	+0.02	[-0.14, +0.19]	spans	spans
Qwen 14B	OilRig	+0.134	+0.222	[+0.080, +0.267]	[-0.007, +0.451]	not clear	not clear	+1.09	+2.53	[+1.17, +3.37]	[+0.02, +5.04]	not clear	above	-0.06	+0.08	[-0.25, +0.45]	spans	spans
Qwen 14B	Sandworm	+0.222	+0.228	[-0.045, +0.235]	[-0.051, +0.507]	not clear	not clear	+1.10	+1.22	[+0.12, +1.44]	[+0.05, +2.39]	above	above	+0.25	+0.31	[-0.09, +0.67]	spans	spans
Qwen 14B	menuPass	+0.060	+0.060	[-0.077, +0.102]	[-0.221, +0.341]	not clear	not clear	+0.23	+0.23	[-1.30, +1.17]	[-4.40, +4.86]	not clear	not clear	+0.68	+0.68	[+0.28, +1.11]	above	above
Qwen 14B sampled	APT29	+0.160	+0.160	[-0.019, +0.200]	[-0.013, +0.334]	not clear	not clear	-0.15	-0.15	[-0.94, +0.67]	[-1.15, +0.85]	not clear	not clear	+0.00	+0.00	[-0.17, +0.18]	spans	spans
Qwen 14B sampled	OilRig	+0.154	+0.190	[+0.015, +0.251]	[-0.108, +0.489]	not clear	not clear	+1.23	+2.10	[+0.20, +2.90]	[-1.18, +5.39]	not clear	not clear	-0.10	-0.04	[-0.29, +0.24]	spans	spans
Qwen 14B sampled	Sandworm	+0.130	+0.128	[-0.017, +0.173]	[+0.067, +0.188]	not clear	not clear	+0.49	+0.52	[-0.37, +1.23]	[-0.41, +1.45]	not clear	not clear	+0.06	+0.07	[-0.08, +0.25]	spans	spans
Qwen 14B sampled	menuPass	+0.019	+0.019	[-0.133, +0.212]	[-0.328, +0.367]	not clear	not clear	-0.54	-0.54	[-3.10, +5.63]	[-7.98, +6.90]	not clear	not clear	+0.51	+0.51	[+0.09, +0.97]	above	above
Ministral 14B	APT29	+0.233	+0.233	[-0.009, +0.270]	[+0.019, +0.448]	not clear	not clear	+0.95	+0.95	[-0.27, +1.52]	[-0.07, +1.97]	not clear	not clear	+0.17	+0.17	[-0.04, +0.40]	spans	spans
Ministral 14B	OilRig	+0.375	+0.356	[+0.250, +0.381]	[+0.222, +0.489]	above	above	+3.31	+3.68	[+3.57, +4.25]	[+3.08, +4.27]	above	above	+0.56	+0.65	[+0.25, +1.04]	above	above
Ministral 14B	Sandworm	+0.049	+0.051	[-0.045, +0.062]	[-0.020, +0.123]	not clear	not clear	+0.40	+0.44	[+0.12, +3.32]	[-0.60, +1.48]	not clear	not clear	-0.03	+0.00	[-0.38, +0.40]	spans	spans
Ministral 14B	menuPass	+0.044	+0.044	[-0.105, +0.178]	[-0.028, +0.117]	not clear	not clear	+0.00	+0.00	[-2.30, +5.06]	[-2.13, +2.13]	not clear	not clear	+0.72	+0.72	[-0.32, +1.59]	spans	spans
Ministral 14B sampled	APT29	+0.143	+0.143	[-0.069, +0.197]	[-0.007, +0.293]	not clear	not clear	-0.08	-0.08	[-1.07, +0.57]	[-0.66, +0.51]	not clear	not clear	+0.12	+0.12	[-0.07, +0.32]	spans	spans
Ministral 14B sampled	OilRig	+0.234	+0.259	[+0.188, +0.311]	[+0.152, +0.366]	above	above	+1.99	+2.69	[+2.20, +4.06]	[+0.94, +4.43]	above	above	+0.26	+0.40	[+0.01, +0.81]	spans	above
Ministral 14B sampled	Sandworm	+0.161	+0.169	[-0.048, +0.242]	[-0.136, +0.473]	not clear	not clear	+1.04	+1.14	[-0.16, +2.30]	[-0.89, +3.18]	not clear	not clear	+0.07	+0.11	[-0.23, +0.47]	spans	spans
Ministral 14B sampled	menuPass	+0.025	+0.025	[-0.149, +0.142]	[-0.089, +0.138]	not clear	not clear	-0.54	-0.54	[-3.11, +4.06]	[-3.75, +2.67]	not clear	not clear	+0.64	+0.64	[+0.03, +1.31]	above	above
Ministral 3B	APT29	+0.133	+0.133	[+0.000, +0.157]	[+0.001, +0.266]	not clear	not clear	–	–	–	–	undefined	undefined	-0.24	-0.24	[-0.41, -0.04]	below	below
Ministral 3B	OilRig	+0.204	+0.217	[+0.065, +0.278]	[+0.016, +0.419]	above	above	–	–	–	–	undefined	undefined	+0.08	+0.15	[-0.31, +0.64]	spans	spans
Ministral 3B	Sandworm	+0.162	+0.171	[+0.067, +0.200]	[+0.007, +0.336]	above	above	–	–	–	–	undefined	undefined	+0.01	+0.04	[-0.14, +0.25]	spans	spans
Ministral 3B	menuPass	+0.229	+0.229	[+0.083, +0.308]	[-0.038, +0.495]	not clear	not clear	–	–	–	–	undefined	undefined	+0.89	+0.89	[+0.30, +1.42]	above	above

3 Tactic mapping

Table S15: SigmaHQ r2026-07-01 (`rules/windows`) tags and the frozen tactic each maps to. Release: rules in the release carrying the tag; Mapper: rules in the compiled mapper (2,017 rules); Frozen v3: rules in the frozen subset evaluable on both eCAR and Sysmon (707); Decides (v3): v3 rules in which the tag is the first tactic tag, which sets an event's tactic. *: the release has no Defense Evasion tag, so these two tags map to it.

Sigma tag	Frozen tactic	Release	Mapper	Frozen v3	Decides (v3)
`attack.reconnaissance`	Reconnaissance	12	7	0	0
`attack.resource-development`	Resource Development	22	20	11	10
`attack.initial-access`	Initial Access	63	58	31	20
`attack.execution`	Execution	693	553	182	121
`attack.persistence`	Persistence	528	470	155	84
`attack.privilege-escalation`	Privilege Escalation	480	440	121	90
`attack.defense-impairment`*	Defense Evasion	302	245	63	38
`attack.stealth`*	Defense Evasion	800	701	255	163
`attack.credential-access`	Credential Access	256	204	66	53
`attack.discovery`	Discovery	165	127	36	28
`attack.lateral-movement`	Lateral Movement	115	95	26	13
`attack.collection`	Collection	66	55	21	18
`attack.command-and-control`	Command and Control	185	163	65	53
`attack.exfiltration`	Exfiltration	47	36	7	7
`attack.impact`	Impact	61	47	9	9
`attack.ds0005`	dropped (attack tag is not a tactic)	1			
`attack.gNNNN / attack.sNNNN`	dropped (group/software tag)	62			
`attack.tNNNN[.NNN]`	dropped (technique tag)	2896			
`non-ATT&CK tag (car.*)`	dropped (not an attack tag)	61			
`non-ATT&CK tag (cve.*)`	dropped (not an attack tag)	12			
`non-ATT&CK tag (detection.*)`	dropped (not an attack tag)	1			
`non-ATT&CK tag (stp.*)`	dropped (not an attack tag)	6			

4 Resampling and chance

Table S16: F1 ÷ chance computed exactly (hypergeometric over the 182 edges) and checked against the identity F1 ÷ chance = precision ÷ ($|E_g|/182$) for every kernel.

Kernel	Plan	edge-F1	exact chance F1	TP/E[TP]	precision/(|Egl/182)	exact percentile
R-F	APT29	0.371	0.1867	1.989	1.989	99.8
R-F	OilRig	0.396	0.1226	3.232	3.232	100.0
R-F	Sandworm	0.372	0.1406	2.648	2.648	99.9
R-F	menuPass	0.300	0.0518	5.783	5.783	99.9
R-G	APT29	0.450	0.2034	2.213	2.213	100.0
R-G	OilRig	0.341	0.1257	2.717	2.717	100.0
R-G	Sandworm	0.331	0.1489	2.223	2.223	99.7
R-G	menuPass	0.181	0.0547	3.301	3.301	97.5
R-Q	APT29	0.358	0.1791	2.000	2.000	99.7
R-Q	OilRig	0.205	0.1014	2.022	2.022	92.6
R-Q	Sandworm	0.340	0.1393	2.438	2.438	99.8
R-Q	menuPass	0.270	0.0552	4.892	4.892	100.0
R-Q5	APT29	0.359	0.1857	1.985	1.985	97.5
R-Q5	OilRig	0.199	0.1034	1.875	1.875	83.0
R-Q5	Sandworm	0.236	0.1399	1.706	1.706	86.6
R-Q5	menuPass	0.239	0.0546	4.441	4.441	99.5
R-M	APT29	0.333	0.1641	2.031	2.031	99.4
R-M	OilRig	0.375	0.1133	3.309	3.309	100.0
R-M	Sandworm	0.167	0.0962	1.733	1.733	84.1
R-M	menuPass	0.267	0.0527	5.056	5.056	99.8
R-M5	APT29	0.311	0.1828	1.708	1.708	96.0
R-M5	OilRig	0.250	0.1131	2.211	2.211	97.0
R-M5	Sandworm	0.265	0.1070	2.476	2.476	96.0
R-M5	menuPass	0.290	0.0529	5.498	5.498	99.6
R-M3	APT29	0.133	0.1641	0.812	0.812	32.6
R-M3	OilRig	0.204	0.1144	1.784	1.784	91.9
R-M3	Sandworm	0.162	0.0998	1.625	1.625	81.5
R-M3	menuPass	0.229	0.0546	4.184	4.184	99.7
B-F	APT29	0.234	0.1191	1.959	1.959	94.3
B-F	OilRig	0.116	0.0835	1.374	1.374	66.9
B-F	Sandworm	0.183	0.0988	1.847	1.847	81.9
B-F	menuPass	0.495	0.0455	10.867	10.867	100.0
B-G	APT29	0.177	0.1073	1.649	1.649	75.9
B-G	OilRig	0.149	0.0617	2.373	2.373	78.5
B-G	Sandworm	0.181	0.0743	2.448	2.448	91.3
B-G	menuPass	0.323	0.0370	8.869	8.869	99.0
B-Q	APT29	0.200	0.0923	2.167	2.167	94.3
B-Q	OilRig	0.071	0.0765	0.933	0.933	51.1
B-Q	Sandworm	0.118	0.0882	1.333	1.333	68.3
B-Q	menuPass	0.211	0.0451	4.667	4.667	96.7
B-M	APT29	0.100	0.0923	1.083	1.083	57.3
B-M	OilRig	0.000	0.0733	0.000	0.000	17.2
B-M	Sandworm	0.118	0.0882	1.333	1.333	68.3
B-M	menuPass	0.222	0.0440	5.056	5.056	97.2
ATTK	APT29	0.238	0.2538	0.936	0.936	27.4
ATTK	OilRig	0.204	0.1480	1.379	1.379	99.7
ATTK	Sandworm	0.207	0.2069	1.000	1.000	50.0
ATTK	menuPass	0.069	0.0625	1.103	1.103	60.5

Table S17: Effective smoothing of Eq. 1 at $\alpha = 0.5$ per kernel: number of reports n, samples, largest row total, maximum attainable and fitted row concentration, in probability and in bits above uniform. The ATTK kernel has no report-count cap.

Kernel	Actor	n	samples	max row total	max attainable P	cap bits	fitted max P	fitted bits
R-F	APT29	7	5	10	0.556	2.85	0.333	2.12
R-G	APT29	7	5	12	0.556	2.85	0.333	2.12
R-Q	APT29	7	1	16	0.556	2.85	0.368	2.26
R-Q5	APT29	7	5	24	0.556	2.85	0.478	2.64
R-M	APT29	7	1	7	0.556	2.85	0.360	2.23
R-M5	APT29	7	5	14	0.556	2.85	0.429	2.48
R-M3	APT29	7	1	6	0.556	2.85	0.263	1.77
B-F	APT29	1	5	3	0.200	1.38	0.200	1.38
B-G	APT29	1	5	3	0.200	1.38	0.200	1.38
B-Q	APT29	1	1	1	0.200	1.38	0.200	1.38
B-M	APT29	1	1	2	0.200	1.38	0.200	1.38
ATTK	APT29	–	1	1110	0.203	1.40	0.202	1.39
R-F	OilRig	5	5	17	0.478	2.64	0.368	2.26
R-G	OilRig	5	5	17	0.478	2.64	0.440	2.52
R-Q	OilRig	5	1	7	0.478	2.64	0.478	2.64
R-Q5	OilRig	5	5	13	0.478	2.64	0.478	2.64
R-M	OilRig	5	1	12	0.478	2.64	0.360	2.23
R-M5	OilRig	5	5	16	0.478	2.64	0.391	2.35
R-M3	OilRig	5	1	9	0.478	2.64	0.333	2.12
B-F	OilRig	1	5	3	0.200	1.38	0.200	1.38
B-G	OilRig	1	5	4	0.200	1.38	0.200	1.38
B-Q	OilRig	1	1	1	0.200	1.38	0.200	1.38
B-M	OilRig	1	1	2	0.200	1.38	0.200	1.38
ATTK	OilRig	–	1	1140	0.195	1.34	0.194	1.34
R-F	Sandworm	6	5	10	0.520	2.76	0.368	2.26
R-G	Sandworm	6	5	10	0.520	2.76	0.304	1.98
R-Q	Sandworm	6	1	10	0.520	2.76	0.217	1.50
R-Q5	Sandworm	6	5	12	0.520	2.76	0.368	2.26
R-M	Sandworm	6	1	5	0.520	2.76	0.294	1.93
R-M5	Sandworm	6	5	5	0.520	2.76	0.304	1.98
R-M3	Sandworm	6	1	4	0.520	2.76	0.200	1.38
B-F	Sandworm	1	5	3	0.200	1.38	0.200	1.38
B-G	Sandworm	1	5	4	0.200	1.38	0.200	1.38
B-Q	Sandworm	1	1	1	0.200	1.38	0.200	1.38
B-M	Sandworm	1	1	1	0.200	1.38	0.200	1.38
ATTK	Sandworm	–	1	948	0.146	0.93	0.146	0.92
R-F	menuPass	5	5	7	0.478	2.64	0.391	2.35
R-G	menuPass	5	5	8	0.478	2.64	0.333	2.12
R-Q	menuPass	5	1	12	0.478	2.64	0.304	1.98
R-Q5	menuPass	5	5	13	0.478	2.64	0.429	2.48
R-M	menuPass	5	1	6	0.478	2.64	0.304	1.98
R-M5	menuPass	5	5	7	0.478	2.64	0.333	2.12
R-M3	menuPass	5	1	10	0.478	2.64	0.294	1.93
B-F	menuPass	1	5	3	0.200	1.38	0.200	1.38
B-G	menuPass	1	5	2	0.200	1.38	0.200	1.38
B-Q	menuPass	1	1	1	0.200	1.38	0.200	1.38
B-M	menuPass	1	1	1	0.200	1.38	0.200	1.38
ATTK	menuPass	–	1	516	0.255	1.73	0.252	1.71

Table S18: R−B gaps for every model (exact chance; 10,000 resamples). For each of ΔF1, Δenrichment (F1 ÷ chance) and ΔIG: the report-bootstrap and leave-one-report-out jackknife 95% intervals and whether the gap is clear (both exclude zero on the same side). For ΔIG, the paired interval from resampling plan transitions (i.i.d. and blocks of 2/3/5 within scenarios) is also shown. B samples: number of no-report generations averaged. Ministral 3B is shown for completeness but not interpreted: without a report it returned an empty list in 19 of 24 calls, so its gaps are measured against a nearly empty kernel.

Model	Actor	B samples	ΔF1	report boot 95%	jackknife 95%	clear	Δenrichment	report boot 95%	jackknife 95%	clear	ΔIG	transition i.i.d. 95%	block b=2/3/5	ΔIG report boot 95%	ΔIG jackknife 95%	ΔIG report-level clear
Claude Opus	APT29	5	+0.137	[-0.098, +0.177]	[-0.065, +0.339]	not clear	+0.03	[-0.85, +0.44]	[-0.87, +0.93]	not clear	+0.06	[-0.15, +0.27]	spans/spans/spans	[-0.27, +0.11]	[-0.28, +0.39]	not clear
Claude Opus	OilRig	5	+0.280	[+0.185, +0.329]	[+0.181, +0.380]	above	+1.86	[+1.29, +2.98]	[+0.42, +3.30]	above	+0.61	[+0.18, +1.02]	above/above/above	[+0.07, +0.69]	[+0.28, +0.94]	above
Claude Opus	Sandworm	5	+0.189	[-0.012, +0.255]	[-0.009, +0.386]	not clear	+0.80	[-0.05, +1.79]	[-0.30, +1.90]	not clear	+0.33	[-0.07, +0.72]	spans/spans/above *	[-0.11, +0.46]	[-0.27, +0.93]	not clear
Claude Opus	menuPass	5	-0.195	[-0.491, -0.062]	[-0.544, +0.154]	not clear	-5.08	[-10.73, -1.52]	[-11.98, +1.81]	not clear	-0.09	[-0.51, +0.26]	spans/spans/spans	[-1.22, +0.37]	[-1.50, +1.32]	not clear
GPT-6 Astra	APT29	5	+0.273	[+0.023, +0.342]	[+0.021, +0.526]	above	+0.56	[-0.41, +1.33]	[-0.51, +1.64]	not clear	+0.30	[+0.06, +0.53]	above/above/above	[-0.19, +0.40]	[-0.15, +0.75]	not clear
GPT-6 Astra	OilRig	5	+0.193	[+0.115, +0.297]	[+0.098, +0.288]	above	+0.34	[-0.56, +2.16]	[-0.80, +1.49]	not clear	+0.50	[-0.02, +1.00]	spans/spans/above *	[+0.02, +0.66]	[+0.10, +0.90]	above
GPT-6 Astra	Sandworm	5	+0.150	[-0.092, +0.217]	[-0.280, +0.580]	not clear	-0.22	[-1.69, +0.64]	[-2.73, +2.28]	not clear	+0.17	[-0.14, +0.46]	spans/spans/spans	[-0.25, +0.32]	[-0.53, +0.86]	not clear
GPT-6 Astra	menuPass	5	-0.143	[-0.307, -0.082]	[-0.223, -0.062]	below	-5.57	[-8.85, -3.47]	[-6.87, -4.26]	below	-0.01	[-0.92, +0.74]	spans/spans/below *	[-0.68, +0.20]	[-0.63, +0.62]	not clear
Qwen 14B	APT29	1	+0.158	[-0.067, +0.207]	[-0.213, +0.530]	not clear	-0.17	[-1.02, +0.43]	[-1.83, +1.50]	not clear	+0.02	[-0.14, +0.19]	spans/spans/spans	[-0.31, +0.10]	[-0.36, +0.40]	not clear
Qwen 14B	OilRig	1	+0.134	[+0.000, +0.171]	[-0.079, +0.346]	not clear	+1.09	[+0.00, +1.76]	[-0.92, +3.10]	not clear	-0.06	[-0.41, +0.29]	spans/spans/spans	[-0.31, +0.08]	[-0.38, +0.27]	not clear
Qwen 14B	Sandworm	1	+0.222	[-0.044, +0.229]	[-0.053, +0.497]	not clear	+1.10	[+0.11, +1.30]	[+0.05, +2.16]	above	+0.25	[-0.11, +0.59]	spans/spans/spans	[-0.05, +0.25]	[-0.15, +0.64]	not clear
Qwen 14B	menuPass	1	+0.060	[-0.077, +0.102]	[-0.221, +0.341]	not clear	+0.23	[-1.30, +1.17]	[-4.40, +4.86]	not clear	+0.68	[+0.28, +1.11]	above/above/above	[-0.16, +1.07]	[-0.77, +2.12]	not clear
Qwen 14B sampled	APT29	5	+0.160	[-0.019, +0.200]	[-0.013, +0.334]	not clear	-0.15	[-0.94, +0.67]	[-1.15, +0.85]	not clear	+0.00	[-0.17, +0.18]	spans/spans/spans	[-0.35, +0.10]	[-0.31, +0.32]	not clear
Qwen 14B sampled	OilRig	5	+0.154	[-0.034, +0.244]	[-0.144, +0.452]	not clear	+1.23	[-0.52, +2.18]	[-1.52, +3.98]	not clear	-0.10	[-0.33, +0.15]	spans/spans/spans	[-0.45, +0.20]	[-0.57, +0.38]	not clear
Qwen 14B sampled	Sandworm	5	+0.130	[-0.016, +0.174]	[+0.076, +0.183]	not clear	+0.49	[-0.32, +1.11]	[-0.30, +1.29]	not clear	+0.06	[-0.08, +0.22]	spans/spans/spans	[-0.20, +0.14]	[-0.16, +0.28]	not clear
Qwen 14B sampled	menuPass	5	+0.019	[-0.133, +0.212]	[-0.328, +0.367]	not clear	-0.54	[-3.10, +5.63]	[-7.98, +6.90]	not clear	+0.51	[+0.09, +0.97]	above/above/above	[-0.28, +0.84]	[-0.65, +1.67]	not clear
Ministral 14B	APT29	1	+0.233	[-0.009, +0.270]	[+0.019, +0.448]	not clear	+0.95	[-0.27, +1.52]	[-0.07, +1.97]	not clear	+0.17	[-0.04, +0.40]	spans/spans/spans	[-0.17, +0.22]	[-0.16, +0.51]	not clear
Ministral 14B	OilRig	1	+0.375	[+0.222, +0.400]	[+0.147, +0.603]	above	+3.31	[+2.85, +3.88]	[+2.05, +4.57]	above	+0.56	[+0.22, +0.91]	above/above/above	[+0.16, +0.64]	[+0.06, +1.06]	above
Ministral 14B	Sandworm	1	+0.049	[-0.044, +0.059]	[-0.022, +0.120]	not clear	+0.40	[+0.11, +3.00]	[-0.54, +1.34]	not clear	-0.03	[-0.38, +0.34]	spans/spans/spans	[-0.14, +0.05]	[-0.12, +0.07]	not clear
Ministral 14B	menuPass	1	+0.044	[-0.105, +0.178]	[-0.028, +0.117]	not clear	+0.00	[-2.30, +5.06]	[-2.13, +2.13]	not clear	+0.72	[-0.32, +1.59]	spans/spans/above *	[-0.12, +0.97]	[-0.09, +1.53]	not clear
Ministral 14B sampled	APT29	5	+0.143	[-0.069, +0.197]	[-0.007, +0.293]	not clear	-0.08	[-1.07, +0.57]	[-0.66, +0.51]	not clear	+0.12	[-0.07, +0.32]	spans/spans/spans	[-0.18, +0.18]	[-0.17, +0.41]	not clear
Ministral 14B sampled	OilRig	5	+0.234	[+0.157, +0.274]	[+0.158, +0.311]	above	+1.99	[+1.40, +3.09]	[+0.74, +3.24]	above	+0.26	[-0.08, +0.64]	spans/spans/above *	[-0.10, +0.41]	[-0.16, +0.69]	not clear
Ministral 14B sampled	Sandworm	5	+0.161	[-0.046, +0.229]	[-0.131, +0.453]	not clear	+1.04	[-0.15, +2.08]	[-0.80, +2.88]	not clear	+0.07	[-0.25, +0.39]	spans/spans/spans	[-0.14, +0.14]	[-0.16, +0.29]	not clear
Ministral 14B sampled	menuPass	5	+0.025	[-0.149, +0.142]	[-0.089, +0.138]	not clear	-0.54	[-3.11, +4.06]	[-3.75, +2.67]	not clear	+0.64	[+0.03, +1.31]	spans/above/above *	[-0.25, +1.08]	[-0.29, +1.58]	not clear
Ministral 3B	APT29	1	+0.133	[+0.000, +0.157]	[+0.001, +0.266]	not clear	–	–	–	undefined	-0.24	[-0.41, -0.04]	below/below/below	[-0.44, -0.08]	[-0.44, -0.04]	below
Ministral 3B	OilRig	1	+0.204	[+0.059, +0.256]	[+0.016, +0.392]	above	–	–	–	undefined	+0.08	[-0.32, +0.51]	spans/spans/spans	[-0.35, +0.21]	[-0.34, +0.50]	not clear
Ministral 3B	Sandworm	1	+0.162	[+0.062, +0.187]	[+0.008, +0.317]	above	–	–	–	undefined	+0.01	[-0.15, +0.21]	spans/spans/spans	[-0.13, +0.08]	[-0.15, +0.18]	not clear
Ministral 3B	menuPass	1	+0.229	[+0.083, +0.308]	[-0.038, +0.495]	not clear	–	–	–	undefined	+0.89	[+0.30, +1.42]	above/above/above	[+0.06, +1.19]	[-0.22, +2.01]	not clear

Table S19: Information gain over uniform (main Table 3) with i.i.d. 95% intervals (10,000 resamples), status under segment-respecting blocks of 2, 3 and 5, and the draft's earlier block marks (circular, crossing scenario boundaries), kept for comparison.

Kernel	Plan	IG	i.i.d. 95%	block b=2	block b=3	block b=5	status i.i.d. → b=2/3/5	* (adopted)	old circular b=2/3/5 (draft, 2,000)	old * (draft)
R-F	APT29	+.11	[-0.13, +0.37]	[-0.13, +0.40]	[-0.14, +0.40]	[-0.13, +0.40]	spans → spans/spans/spans		spans/spans/spans	
R-F	OilRig	+.62	[+0.17, +1.04]	[+0.17, +0.96]	[+0.22, +0.91]	[+0.30, +0.93]	above → above/above/above		above/above/above	
R-F	Sandworm	+.31	[-0.05, +0.68]	[-0.04, +0.71]	[-0.06, +0.67]	[-0.06, +0.74]	spans → spans/spans/spans		spans/spans/spans	
R-F	menuPass	+.91	[+0.33, +1.49]	[+0.28, +1.52]	[+0.60, +1.48]	[+0.91, +1.13]	above → above/above/above		above/above/above	
R-G	APT29	+.30	[+0.05, +0.55]	[+0.08, +0.54]	[+0.07, +0.58]	[+0.08, +0.58]	above → above/above/above		above/above/above	
R-G	OilRig	+.55	[-0.01, +1.08]	[-0.07, +1.00]	[-0.05, +0.98]	[+0.02, +1.05]	spans → spans/spans/above	*	spans/spans/above	*
R-G	Sandworm	+.18	[-0.15, +0.52]	[-0.21, +0.54]	[-0.26, +0.56]	[-0.31, +0.62]	spans → spans/spans/spans		spans/spans/spans	
R-G	menuPass	+.39	[-0.31, +1.13]	[-0.33, +1.35]	[+0.07, +1.05]	[+0.21, +0.39]	spans → spans/above/above	*	spans/spans/above	*
R-Q	APT29	+.12	[-0.12, +0.37]	[-0.11, +0.36]	[-0.10, +0.38]	[-0.09, +0.40]	spans → spans/spans/spans		spans/spans/spans	
R-Q	OilRig	-.17	[-0.45, +0.14]	[-0.43, -0.01]	[-0.41, -0.00]	[-0.40, -0.02]	spans → below/below/below	*	spans/spans/spans	
R-Q	Sandworm	+.17	[-0.11, +0.46]	[-0.11, +0.49]	[-0.10, +0.54]	[-0.06, +0.58]	spans → spans/spans/spans		spans/spans/spans	
R-Q	menuPass	+1.00	[+0.55, +1.37]	[+0.60, +1.32]	[+0.85, +1.24]	[+0.88, +1.07]	above → above/above/above		above/above/above	
R-Q5	APT29	+.10	[-0.15, +0.35]	[-0.15, +0.34]	[-0.16, +0.34]	[-0.14, +0.37]	spans → spans/spans/spans		spans/spans/spans	
R-Q5	OilRig	-.21	[-0.42, +0.03]	[-0.41, -0.03]	[-0.40, -0.01]	[-0.38, -0.03]	spans → below/below/below	*	below/spans/below	*
R-Q5	Sandworm	-.02	[-0.18, +0.17]	[-0.20, +0.20]	[-0.20, +0.24]	[-0.20, +0.25]	spans → spans/spans/spans		spans/spans/spans	
R-Q5	menuPass	+.86	[+0.35, +1.36]	[+0.18, +1.35]	[+0.40, +1.33]	[+0.84, +0.88]	above → above/above/above		above/above/above	
R-M	APT29	+.16	[-0.10, +0.42]	[-0.06, +0.39]	[-0.08, +0.41]	[-0.07, +0.43]	spans → spans/spans/spans		spans/spans/spans	
R-M	OilRig	+.37	[+0.07, +0.69]	[+0.06, +0.69]	[+0.12, +0.71]	[+0.16, +0.72]	above → above/above/above		above/above/above	
R-M	Sandworm	-.04	[-0.26, +0.24]	[-0.28, +0.33]	[-0.27, +0.37]	[-0.25, +0.39]	spans → spans/spans/spans		spans/spans/spans	
R-M	menuPass	+1.04	[+0.35, +1.73]	[+0.07, +1.64]	[+0.28, +1.49]	[+0.96, +1.11]	above → above/above/above		above/above/above	
R-M5	APT29	+.13	[-0.10, +0.38]	[-0.08, +0.39]	[-0.09, +0.42]	[-0.11, +0.46]	spans → spans/spans/spans		spans/spans/spans	
R-M5	OilRig	+.07	[-0.28, +0.42]	[-0.27, +0.42]	[-0.23, +0.38]	[-0.10, +0.35]	spans → spans/spans/spans		spans/spans/spans	
R-M5	Sandworm	+.06	[-0.17, +0.31]	[-0.17, +0.37]	[-0.16, +0.41]	[-0.15, +0.42]	spans → spans/spans/spans		spans/spans/spans	
R-M5	menuPass	+1.07	[+0.54, +1.57]	[+0.58, +1.42]	[+0.50, +1.21]	[+0.99, +1.14]	above → above/above/above		above/above/above	
R-M3	APT29	-.24	[-0.41, -0.04]	[-0.39, -0.05]	[-0.41, -0.05]	[-0.41, -0.04]	below → below/below/below		below/below/below	
R-M3	OilRig	+.08	[-0.32, +0.51]	[-0.37, +0.52]	[-0.37, +0.44]	[-0.41, +0.22]	spans → spans/spans/spans		spans/spans/spans	
R-M3	Sandworm	+.01	[-0.15, +0.21]	[-0.15, +0.28]	[-0.16, +0.32]	[-0.16, +0.33]	spans → spans/spans/spans		spans/spans/spans	
R-M3	menuPass	+.89	[+0.30, +1.42]	[+0.53, +1.09]	[+0.73, +0.99]	[+0.56, +0.89]	above → above/above/above		above/above/above	
B-F	APT29	+.06	[-0.08, +0.22]	[-0.09, +0.22]	[-0.10, +0.23]	[-0.09, +0.23]	spans → spans/spans/spans		spans/spans/spans	
B-F	OilRig	+.01	[-0.18, +0.25]	[-0.22, +0.23]	[-0.24, +0.20]	[-0.22, +0.14]	spans → spans/spans/spans		spans/spans/spans	
B-F	Sandworm	-.02	[-0.17, +0.16]	[-0.17, +0.18]	[-0.17, +0.22]	[-0.17, +0.23]	spans → spans/spans/spans		spans/spans/spans	
B-F	menuPass	+1.01	[+0.57, +1.33]	[+0.50, +1.31]	[+0.74, +1.31]	[+1.01, +1.09]	above → above/above/above		above/above/above	
B-G	APT29	+.00	[-0.13, +0.16]	[-0.13, +0.16]	[-0.13, +0.14]	[-0.13, +0.14]	spans → spans/spans/spans		spans/spans/spans	
B-G	OilRig	+.05	[-0.07, +0.21]	[-0.07, +0.24]	[-0.07, +0.24]	[-0.06, +0.16]	spans → spans/spans/spans		spans/spans/spans	
B-G	Sandworm	+.01	[-0.14, +0.20]	[-0.16, +0.27]	[-0.16, +0.31]	[-0.16, +0.34]	spans → spans/spans/spans		spans/spans/spans	
B-G	menuPass	+.40	[-0.12, +0.93]	[+0.13, +0.71]	[+0.39, +0.93]	[+0.39, +0.67]	spans → above/above/above	*	above/spans/above	*
B-Q	APT29	+.10	[-0.06, +0.29]	[-0.04, +0.30]	[-0.03, +0.30]	[-0.01, +0.31]	spans → spans/spans/spans		spans/spans/spans	
B-Q	OilRig	-.11	[-0.21, +0.07]	[-0.21, +0.07]	[-0.21, +0.07]	[-0.21, +0.07]	spans → spans/spans/spans		spans/spans/spans	
B-Q	Sandworm	-.07	[-0.21, +0.12]	[-0.21, +0.11]	[-0.21, +0.11]	[-0.21, +0.06]	spans → spans/spans/spans		spans/spans/spans	
B-Q	menuPass	+.32	[-0.21, +0.85]	[-0.21, +0.59]	[+0.32, +0.32]	[+0.06, +0.32]	spans → spans/above/above	*	spans/above/above	*
B-M	APT29	-.01	[-0.11, +0.11]	[-0.11, +0.12]	[-0.11, +0.13]	[-0.12, +0.14]	spans → spans/spans/spans		spans/spans/spans	
B-M	OilRig	-.19	[-0.26, -0.12]	[-0.26, -0.13]	[-0.25, -0.14]	[-0.24, -0.15]	below → below/below/below		below/below/below	
B-M	Sandworm	-.01	[-0.20, +0.19]	[-0.19, +0.18]	[-0.14, +0.17]	[-0.14, +0.17]	spans → spans/spans/spans		spans/spans/spans	
B-M	menuPass	+.32	[-0.21, +0.85]	[-0.21, +0.59]	[-0.21, +0.85]	[+0.06, +0.32]	spans → spans/spans/above	*	spans/spans/above	*
ATTK	APT29	-.61	[-1.27, -0.02]	[-1.31, +0.01]	[-1.39, +0.08]	[-1.30, +0.19]	below → spans/spans/spans	*	spans/spans/spans	*
ATTK	OilRig	-.05	[-0.72, +0.55]	[-0.64, +0.44]	[-0.47, +0.48]	[-0.41, +0.44]	spans → spans/spans/spans		spans/spans/spans	
ATTK	Sandworm	-.30	[-0.59, -0.06]	[-0.55, -0.06]	[-0.53, -0.07]	[-0.47, -0.12]	below → below/below/below		below/below/below	
ATTK	menuPass	-.92	[-2.99, +0.38]	[-2.78, +0.15]	[-2.07, +0.23]	[-1.00, -0.85]	spans → spans/spans/below	*	spans/spans/spans	

5 Case studies

Table S20: Case study A (OTRF recording of the APT29 emulation; 37 transitions, 22 edges): information gain over uniform with i.i.d. 95% intervals (10,000 resamples) and block status, and support recovery against the executed sequence (exact chance). The executed sequence matches the plan at 2.07× chance.

Kernel	Kernel actor	IG	i.i.d. 95%	status	block b=2/3/5	edge-F1	F1/chance (exact)	pct (exact)
R-F	APT29	-.12	[-0.34, +0.10]	spans	spans/spans/spans	0.233	1.45	85.7
R-F	OilRig	-.10	[-0.38, +0.19]	spans	spans/spans/spans	0.232	1.44	87.6
R-F	Sandworm	-.43	[-0.63, -0.22]	below	below/below/below	0.117	0.81	34.7
R-F	menuPass	-.10	[-0.33, +0.15]	spans	spans/spans/spans	0.182	1.51	78.0
R-G	APT29	-.04	[-0.28, +0.19]	spans	spans/spans/spans	0.308	1.79	99.1
R-G	OilRig	-.33	[-0.60, -0.05]	below	below/spans/below	0.222	1.34	83.0
R-G	Sandworm	-.21	[-0.45, +0.04]	spans	spans/spans/spans	0.166	1.08	59.5
R-G	menuPass	+.00	[-0.29, +0.31]	spans	spans/spans/spans	0.226	1.63	90.0
R-Q	APT29	-.06	[-0.32, +0.21]	spans	spans/spans/spans	0.262	1.70	95.6
R-Q	OilRig	-.10	[-0.36, +0.20]	spans	spans/spans/spans	0.174	1.38	76.7
R-Q	Sandworm	-.01	[-0.22, +0.20]	spans	spans/spans/spans	0.222	1.55	88.6
R-Q	menuPass	-.13	[-0.36, +0.10]	spans	spans/spans/spans	0.189	1.33	77.2
R-Q5	APT29	-.05	[-0.29, +0.19]	spans	spans/spans/spans	0.294	1.88	95.9
R-Q5	OilRig	-.08	[-0.29, +0.17]	spans	spans/spans/spans	0.223	1.74	90.4
R-Q5	Sandworm	-.11	[-0.26, +0.05]	spans	spans/spans/spans	0.195	1.36	77.9
R-Q5	menuPass	-.13	[-0.33, +0.08]	spans	spans/spans/spans	0.194	1.41	78.6
R-M	APT29	-.23	[-0.47, +0.04]	spans	spans/spans/spans	0.148	1.03	54.7
R-M	OilRig	-.20	[-0.40, +0.01]	spans	below/below/below	0.109	0.75	29.9
R-M	Sandworm	-.20	[-0.33, -0.07]	below	below/below/below	0.054	0.55	28.4
R-M	menuPass	-.08	[-0.30, +0.17]	spans	spans/spans/spans	0.174	1.38	76.7
R-M5	APT29	-.06	[-0.27, +0.16]	spans	spans/spans/spans	0.221	1.41	82.4
R-M5	OilRig	-.25	[-0.42, -0.07]	below	below/spans/spans	0.124	0.86	39.4
R-M5	Sandworm	-.18	[-0.31, -0.03]	below	below/below/below	0.069	0.63	30.7
R-M5	menuPass	-.01	[-0.22, +0.21]	spans	spans/spans/spans	0.205	1.60	85.2
R-M3	APT29	-.22	[-0.38, -0.04]	below	below/spans/spans	0.111	0.78	32.2
R-M3	OilRig	-.14	[-0.36, +0.12]	spans	spans/spans/spans	0.143	0.97	49.2
R-M3	Sandworm	+.08	[-0.08, +0.26]	spans	spans/spans/spans	0.158	1.55	79.5
R-M3	menuPass	-.03	[-0.22, +0.19]	spans	spans/spans/spans	0.157	1.14	63.2
B-F	APT29	-.05	[-0.19, +0.10]	spans	spans/spans/spans	0.151	1.41	70.6
B-F	OilRig	+.05	[-0.10, +0.22]	spans	spans/spans/spans	0.158	1.58	79.8
B-F	Sandworm	-.10	[-0.19, +0.03]	spans	spans/spans/spans	0.095	0.95	48.4
B-F	menuPass	+.05	[-0.08, +0.19]	spans	spans/spans/spans	0.113	1.24	64.5
B-G	APT29	-.06	[-0.18, +0.10]	spans	spans/spans/spans	0.108	1.10	53.5
B-G	OilRig	-.04	[-0.12, +0.07]	spans	spans/spans/spans	0.090	1.27	62.9
B-G	Sandworm	-.07	[-0.15, +0.02]	spans	spans/spans/spans	0.100	1.34	64.6
B-G	menuPass	-.00	[-0.10, +0.11]	spans	spans/spans/spans	0.053	0.82	47.7
B-Q	APT29	+.01	[-0.16, +0.18]	spans	spans/spans/spans	0.176	2.07	89.7
B-Q	OilRig	+.01	[-0.16, +0.18]	spans	spans/spans/spans	0.171	1.91	87.4
B-Q	Sandworm	+.01	[-0.16, +0.18]	spans	spans/spans/spans	0.171	1.91	87.4
B-Q	menuPass	+.01	[-0.16, +0.18]	spans	spans/spans/spans	0.171	1.91	87.4
B-M	APT29	+.01	[-0.12, +0.18]	spans	spans/spans/spans	0.118	1.38	69.9
B-M	OilRig	+.01	[-0.12, +0.18]	spans	spans/spans/spans	0.118	1.38	69.9
B-M	Sandworm	+.01	[-0.16, +0.18]	spans	spans/spans/spans	0.171	1.91	87.4
B-M	menuPass	-.05	[-0.17, +0.11]	spans	spans/spans/spans	0.118	1.38	69.9
ATTK	APT29	+.76	[+0.50, +0.98]	above	above/above/above	0.286	1.38	100.0
ATTK	OilRig	+.46	[+0.27, +0.64]	above	above/above/above	0.286	1.38	100.0
ATTK	Sandworm	+.09	[-0.11, +0.29]	spans	spans/spans/spans	0.216	1.00	50.0
ATTK	menuPass	+.33	[+0.05, +0.61]	above	above/above/above	0.333	1.65	100.0

Table S21: Case study B (OpTC red team): AUPRC per run and condition with host-cluster bootstrap 95% intervals, draft (1,000 resamples) vs. adopted (10,000). Point AUPRCs are identical in both.

Run	Condition	AUPRC	95% host-cluster (draft, 1,000)	95% host-cluster (10,000)
as_run	ATTK-all	0.306	[0.23, 0.54]	[0.24, 0.52]
as_run	ATTK/APT29	0.189	[0.13, 0.35]	[0.13, 0.32]
as_run	ATTK/OilRig	0.350	[0.26, 0.63]	[0.26, 0.61]
as_run	ATTK/Sandworm	0.369	[0.27, 0.65]	[0.27, 0.63]
as_run	ATTK/menuPass	0.337	[0.25, 0.57]	[0.26, 0.56]
as_run	R-F/APT29	0.143	[0.11, 0.35]	[0.11, 0.31]
as_run	R-F/OilRig	0.133	[0.10, 0.33]	[0.10, 0.29]
as_run	R-F/Sandworm	0.138	[0.10, 0.35]	[0.10, 0.31]
as_run	R-F/menuPass	0.141	[0.10, 0.36]	[0.10, 0.31]
as_run	R-G/APT29	0.126	[0.09, 0.31]	[0.09, 0.27]
as_run	R-G/OilRig	0.129	[0.09, 0.31]	[0.09, 0.28]
as_run	R-G/Sandworm	0.137	[0.10, 0.34]	[0.10, 0.29]
as_run	R-G/menuPass	0.140	[0.10, 0.35]	[0.10, 0.30]
as_run	R-M/APT29	0.129	[0.09, 0.32]	[0.09, 0.30]
as_run	R-M/OilRig	0.147	[0.11, 0.32]	[0.11, 0.29]
as_run	R-M/Sandworm	0.143	[0.11, 0.32]	[0.11, 0.29]
as_run	R-M/menuPass	0.132	[0.10, 0.32]	[0.10, 0.28]
as_run	R-M5/APT29	0.128	[0.09, 0.32]	[0.09, 0.29]
as_run	R-M5/OilRig	0.145	[0.11, 0.33]	[0.11, 0.29]
as_run	R-M5/Sandworm	0.146	[0.11, 0.33]	[0.11, 0.29]
as_run	R-M5/menuPass	0.136	[0.10, 0.32]	[0.10, 0.29]
as_run	R-Q/APT29	0.129	[0.09, 0.33]	[0.09, 0.29]
as_run	R-Q/OilRig	0.145	[0.11, 0.34]	[0.11, 0.30]
as_run	R-Q/Sandworm	0.159	[0.12, 0.37]	[0.12, 0.33]
as_run	R-Q/menuPass	0.132	[0.10, 0.31]	[0.10, 0.28]
as_run	R-Q5/APT29	0.116	[0.08, 0.28]	[0.09, 0.25]
as_run	R-Q5/OilRig	0.135	[0.10, 0.33]	[0.10, 0.29]
as_run	R-Q5/Sandworm	0.131	[0.10, 0.31]	[0.10, 0.28]
as_run	R-Q5/menuPass	0.130	[0.10, 0.32]	[0.10, 0.28]
as_run	U	0.021	[0.01, 0.10]	[0.01, 0.09]
as_run	Uni	0.147	[0.11, 0.35]	[0.11, 0.31]
as_run	oracle-G	0.111	[0.08, 0.29]	[0.08, 0.28]
as_run	oracle-M	0.135	[0.10, 0.32]	[0.10, 0.30]
as_run	oracle-M5	0.135	[0.10, 0.32]	[0.10, 0.30]
as_run	oracle-Q	0.147	[0.11, 0.35]	[0.11, 0.32]
as_run	oracle-Q5	0.145	[0.11, 0.35]	[0.11, 0.31]
collapsed	ATTK-all	0.285	[0.18, 0.49]	[0.18, 0.45]
collapsed	ATTK/APT29	0.282	[0.18, 0.47]	[0.19, 0.44]
collapsed	ATTK/OilRig	0.285	[0.18, 0.49]	[0.19, 0.44]
collapsed	ATTK/Sandworm	0.193	[0.10, 0.28]	[0.10, 0.27]
collapsed	ATTK/menuPass	0.291	[0.19, 0.49]	[0.19, 0.45]
collapsed	R-F/APT29	0.323	[0.24, 0.44]	[0.25, 0.42]
collapsed	R-F/OilRig	0.420	[0.32, 0.61]	[0.33, 0.58]
collapsed	R-F/Sandworm	0.432	[0.34, 0.60]	[0.35, 0.59]
collapsed	R-F/menuPass	0.278	[0.20, 0.37]	[0.21, 0.36]
collapsed	R-G/APT29	0.398	[0.31, 0.59]	[0.31, 0.58]
collapsed	R-G/OilRig	0.469	[0.38, 0.64]	[0.38, 0.62]
collapsed	R-G/Sandworm	0.179	[0.09, 0.26]	[0.09, 0.26]
collapsed	R-G/menuPass	0.444	[0.35, 0.62]	[0.36, 0.61]
collapsed	R-M/APT29	0.178	[0.08, 0.26]	[0.08, 0.26]
collapsed	R-M/OilRig	0.178	[0.08, 0.27]	[0.08, 0.26]
collapsed	R-M/Sandworm	0.178	[0.08, 0.27]	[0.08, 0.26]
collapsed	R-M/menuPass	0.166	[0.07, 0.25]	[0.07, 0.25]
collapsed	R-M5/APT29	0.173	[0.08, 0.26]	[0.08, 0.26]
collapsed	R-M5/OilRig	0.167	[0.07, 0.25]	[0.07, 0.25]
collapsed	R-M5/Sandworm	0.178	[0.08, 0.27]	[0.08, 0.26]
collapsed	R-M5/menuPass	0.158	[0.07, 0.25]	[0.07, 0.24]
collapsed	R-Q/APT29	0.157	[0.08, 0.24]	[0.08, 0.24]
collapsed	R-Q/OilRig	0.177	[0.08, 0.27]	[0.08, 0.26]
collapsed	R-Q/Sandworm	0.167	[0.07, 0.25]	[0.07, 0.25]
collapsed	R-Q/menuPass	0.144	[0.07, 0.23]	[0.07, 0.23]
collapsed	R-Q5/APT29	0.141	[0.07, 0.22]	[0.07, 0.22]
collapsed	R-Q5/OilRig	0.154	[0.07, 0.24]	[0.07, 0.23]
collapsed	R-Q5/Sandworm	0.149	[0.07, 0.23]	[0.07, 0.23]
collapsed	R-Q5/menuPass	0.153	[0.07, 0.24]	[0.07, 0.23]
collapsed	U	0.021	[0.01, 0.10]	[0.01, 0.09]
collapsed	Uni	0.178	[0.08, 0.27]	[0.08, 0.26]
collapsed	oracle-G	0.175	[0.08, 0.26]	[0.08, 0.26]
collapsed	oracle-M	0.172	[0.08, 0.26]	[0.08, 0.26]
collapsed	oracle-M5	0.172	[0.08, 0.26]	[0.08, 0.26]
collapsed	oracle-Q	0.178	[0.08, 0.27]	[0.08, 0.26]
collapsed	oracle-Q5	0.177	[0.08, 0.27]	[0.08, 0.26]
scripted	i_drop_xx01_xx51_hosts/ATTK/APT29	0.609	[0.47, 0.75]	[0.48, 0.75]
scripted	i_drop_xx01_xx51_hosts/ATTK/OilRig	0.609	[0.47, 0.75]	[0.48, 0.75]
scripted	i_drop_xx01_xx51_hosts/ATTK/Sandworm	0.188	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/ATTK/menuPass	0.609	[0.47, 0.75]	[0.48, 0.75]
scripted	i_drop_xx01_xx51_hosts/R-F/APT29	0.469	[0.38, 0.56]	[0.39, 0.56]
scripted	i_drop_xx01_xx51_hosts/R-F/OilRig	0.680	[0.56, 0.82]	[0.56, 0.81]

Table S21 (continued)

Run	Condition	AUPRC	95% host-cluster (draft, 1,000)	95% host-cluster (10,000)
scripted	i_drop_xx01_xx51_hosts/R-F/Sandworm	0.680	[0.56, 0.82]	[0.56, 0.81]
scripted	i_drop_xx01_xx51_hosts/R-F/menuPass	0.384	[0.29, 0.48]	[0.30, 0.49]
scripted	i_drop_xx01_xx51_hosts/R-G/APT29	0.680	[0.56, 0.82]	[0.56, 0.81]
scripted	i_drop_xx01_xx51_hosts/R-G/OilRig	0.680	[0.56, 0.82]	[0.56, 0.81]
scripted	i_drop_xx01_xx51_hosts/R-G/Sandworm	0.153	[0.05, 0.27]	[0.06, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-G/menuPass	0.680	[0.56, 0.82]	[0.56, 0.81]
scripted	i_drop_xx01_xx51_hosts/R-M/APT29	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M/OilRig	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M/Sandworm	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M/menuPass	0.153	[0.05, 0.27]	[0.06, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-M3/APT29	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M3/OilRig	0.153	[0.05, 0.27]	[0.06, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-M3/Sandworm	0.188	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M3/menuPass	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M5/APT29	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M5/OilRig	0.153	[0.05, 0.27]	[0.06, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-M5/Sandworm	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-M5/menuPass	0.164	[0.06, 0.28]	[0.07, 0.28]
scripted	i_drop_xx01_xx51_hosts/R-Q/APT29	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-Q/OilRig	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/R-Q/Sandworm	0.153	[0.05, 0.27]	[0.06, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-Q/menuPass	0.153	[0.05, 0.27]	[0.06, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-Q5/APT29	0.153	[0.06, 0.27]	[0.07, 0.26]
scripted	i_drop_xx01_xx51_hosts/R-Q5/OilRig	0.160	[0.06, 0.27]	[0.07, 0.27]
scripted	i_drop_xx01_xx51_hosts/R-Q5/Sandworm	0.164	[0.06, 0.28]	[0.07, 0.28]
scripted	i_drop_xx01_xx51_hosts/R-Q5/menuPass	0.164	[0.06, 0.28]	[0.07, 0.28]
scripted	i_drop_xx01_xx51_hosts/U	0.657	[0.53, 0.81]	[0.53, 0.81]
scripted	i_drop_xx01_xx51_hosts/Uni	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/oracle-G	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/oracle-M	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/oracle-M5	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/oracle-Q	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	i_drop_xx01_xx51_hosts/oracle-Q5	0.187	[0.08, 0.32]	[0.08, 0.31]
scripted	ii_drop_scripted_rule_tags/ATTK/APT29	0.277	[0.13, 0.40]	[0.14, 0.40]
scripted	ii_drop_scripted_rule_tags/ATTK/OilRig	0.212	[0.10, 0.32]	[0.10, 0.32]
scripted	ii_drop_scripted_rule_tags/ATTK/Sandworm	0.212	[0.10, 0.32]	[0.10, 0.32]
scripted	ii_drop_scripted_rule_tags/ATTK/menuPass	0.212	[0.10, 0.32]	[0.10, 0.32]
scripted	ii_drop_scripted_rule_tags/R-F/APT29	0.151	[0.08, 0.22]	[0.08, 0.22]
scripted	ii_drop_scripted_rule_tags/R-F/OilRig	0.159	[0.08, 0.24]	[0.09, 0.23]
scripted	ii_drop_scripted_rule_tags/R-F/Sandworm	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/R-F/menuPass	0.186	[0.09, 0.29]	[0.09, 0.29]
scripted	ii_drop_scripted_rule_tags/R-G/APT29	0.194	[0.10, 0.28]	[0.10, 0.28]
scripted	ii_drop_scripted_rule_tags/R-G/OilRig	0.216	[0.11, 0.30]	[0.11, 0.30]
scripted	ii_drop_scripted_rule_tags/R-G/Sandworm	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-G/menuPass	0.268	[0.14, 0.37]	[0.15, 0.37]
scripted	ii_drop_scripted_rule_tags/R-M/APT29	0.190	[0.10, 0.31]	[0.10, 0.31]
scripted	ii_drop_scripted_rule_tags/R-M/OilRig	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M/Sandworm	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/R-M/menuPass	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M3/APT29	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M3/OilRig	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M3/Sandworm	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/R-M3/menuPass	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M5/APT29	0.137	[0.07, 0.23]	[0.07, 0.23]
scripted	ii_drop_scripted_rule_tags/R-M5/OilRig	0.128	[0.06, 0.21]	[0.07, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M5/Sandworm	0.137	[0.07, 0.22]	[0.07, 0.21]
scripted	ii_drop_scripted_rule_tags/R-M5/menuPass	0.128	[0.06, 0.21]	[0.07, 0.21]
scripted	ii_drop_scripted_rule_tags/R-Q/APT29	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/R-Q/OilRig	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-Q/Sandworm	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/R-Q/menuPass	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-Q5/APT29	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-Q5/OilRig	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-Q5/Sandworm	0.124	[0.06, 0.21]	[0.06, 0.21]
scripted	ii_drop_scripted_rule_tags/R-Q5/menuPass	0.128	[0.06, 0.21]	[0.07, 0.21]
scripted	ii_drop_scripted_rule_tags/U	0.662	[0.55, 0.80]	[0.55, 0.80]
scripted	ii_drop_scripted_rule_tags/Uni	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/oracle-G	0.128	[0.06, 0.21]	[0.07, 0.21]
scripted	ii_drop_scripted_rule_tags/oracle-M	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/oracle-M5	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/oracle-Q	0.146	[0.07, 0.23]	[0.07, 0.22]
scripted	ii_drop_scripted_rule_tags/oracle-Q5	0.146	[0.07, 0.23]	[0.07, 0.22]

Table S22: Case study B: number of tactic changes per test window, positive and negative windows.

Windows	Tactic changes	count
positive	0	11
positive	1	7
positive	2	22
positive	3	2
positive	4	2
positive	5	2
negative	0	439
negative	1	54
negative	2	202
negative	3	154
negative	4	1280
negative	5	2253